Elementary Statistics for Social Sciences II

208272 (Part 1)

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Preface

This book covers the first half of 208272 Elementary Statistics for Social Sciences II: the principles of statistical inference, inference about means, proportions and variances, and analysis of variance. All examples use economic contexts. Unless a source is given, the numbers in the examples are hypothetical and are used for illustration only.

How to use this book

Each section of Chapters 2–4 matches one class session and follows the same pattern:

1. **Learning objectives**
2. **Concept**: why we need this method
3. **Formulas and conditions**
4. **Worked example**: a full solution
5. **Your Turn**: a similar problem with space to write your own solution
6. **Reading R output** (from Section 3.2 onward)
7. **Formula summary**
8. **Exercises**
9. **Solutions** to Your Turn and the Exercises

In the PDF, the Your Turn problems have blank space for writing, and the solutions are at the end of each section. On the web, click a solution box to open it. Try the problem first.

Session	Section
1	Chapter 1 Basic Principles of Statistical Inference
2	2.1 Mean, σ known
3	2.2 Mean, σ unknown
4	2.3 Proportion
5	2.4 Variance
6	3.1 Two proportions
7	3.2 P-values and R output; 3.3 Paired means
8	3.4 Two variances
9	3.5 Two independent means
10	4.1 One-way ANOVA: CRD
11	4.2 Multiple comparisons (Tukey); 4.3 RCBD
12	Summary: Choosing a Method

Notation for critical values

The subscript of a critical value is the **area to its left**. This is the same convention as the R functions `qnorm()`, `qt()`, `qchisq()` and `qf()`.

Notation	Meaning	R
$z_{0.975}$	the z value with area 0.975 to its left	`qnorm(0.975)`
$t_{0.975,\,24}$	the t value with 24 df and area 0.975 to its left	`qt(0.975, 24)`
$\chi^2_{0.025,\,10}$	the χ^2 value with 10 df and area 0.025 to its left	`qchisq(0.025, 10)`
$F_{0.975,\,5,\,8}$	the F value with (5, 8) df and area 0.975 to its left	`qf(0.975, 5, 8)`

The statistical tables for this course give the same left-tail areas:

Distribution	Table	Left-tail areas in the table
Standard normal (z)	Table 4	z from 0.00 to 3.49
t	Table 5	0.60, 0.70, 0.80, 0.90, 0.95, 0.975, 0.99, 0.995, 0.999, 0.9995
Chi-square (χ^2)	Table 6	0.0005 to 0.9995
F	Table 7	0.90, 0.95, 0.975, 0.99, 0.995

A value calculated from the sample has the subscript "cal", for example z_{cal}, t_{cal}, χ^2_{cal} and F_{cal}.

The five steps of a hypothesis test

1. **Hypotheses**: state H_0 and H_1.
2. **Level of significance**: state α.
3. **Test statistic**: write the formula, substitute the values, and give the result.
4. **Critical value and critical region** (Chapters 1–2 and Section 3.1), or the **p-value** (from Section 3.2 onward).
5. **Decision and interpretation**.

Writing the conclusion

Decision using the critical region (Chapters 1–2 and Section 3.1)

> t_{cal} = … [falls / does not fall] in the critical region (t_{cal} [<, >] …), so we [reject / accept] H_0.

Decision using the p-value (from Section 3.2 onward)

> p-value = … [<, ≥] $\alpha = 0.05$, so we [reject / accept] H_0.

Interpretation

> That is, … at the 0.05 level of significance.

- If we reject H_0, write H_1 as a sentence.
- If we accept H_0, write H_0 as a sentence.
- For a two-sided test that rejects H_0, also state the direction, using the sample statistic.

Confidence interval

> We are 95% confident that … is between … and ….

1 Basic Principles of Statistical Inference

Learning Objectives

After this chapter you should be able to:

1. Distinguish a **parameter** from a **statistic**.
2. Describe the sampling distribution of $\bar{x}$ and of $\hat{p}$: its mean, its standard error and its shape.
3. Use the Central Limit Theorem to find probabilities about $\bar{x}$ and $\hat{p}$.
4. Explain what a 95% confidence interval means.
5. Write the null and alternative hypotheses for a research question, and tell a one-sided test from a two-sided test.
6. Find critical values and the critical region from the z table (Table 4).
7. Explain Type I and Type II errors in an economic setting.

> Why Does This Matter?
>
> Economists rarely see a whole population. The Ministry of Labour cannot ask every worker in Chiang Mai about their wage, and the Bank of Thailand cannot survey every household about its debt. Instead we take a **sample** and use it to say something about the **population**.
> Two samples from the same population give different answers. Statistical inference tells us how much a sample result can vary, and so how far we can trust it. This chapter builds the ideas used in every later chapter.

1.1 Parameters, Statistics and Sampling Distributions

A **parameter** describes a population. It is a fixed number, but usually unknown. A **statistic** describes a sample. We can calculate it, but it changes from sample to sample.

	Population (parameter)	Sample (statistic)
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Variance	σ^2	s^2
Proportion	p	$\hat{p}$

	Population (parameter)	Sample (statistic)
Size	N	n

The **sampling distribution** of a statistic is the distribution of its values over all possible samples of the same size n.

Example. Figure 1.1 uses a hypothetical population of 100,000 households. Monthly household income is skewed to the right: most households earn a moderate amount and a few earn a great deal. The population mean is μ = 27,054 baht and the standard deviation is σ = 19,543 baht. The other panels show the means of 5,000 random samples of size 5, 30 and 100.

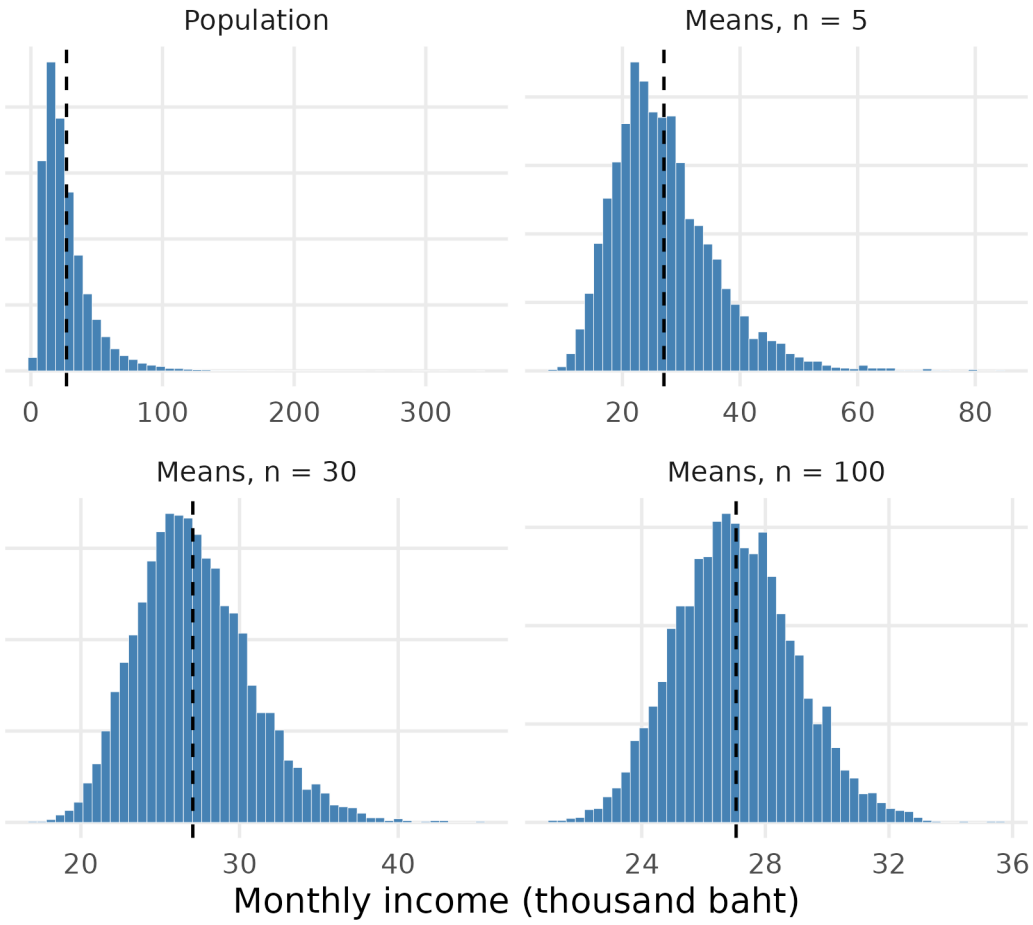


Figure 1.1: Household income in a hypothetical population (top left) and the sampling distribution of the sample mean for n = 5, 30 and 100

Three things happen as n grows:

1. The sample means are centred on the population mean μ (dashed line).
2. The spread of the sample means gets **smaller**.

3. The shape gets closer to a **normal** curve, even though the population is skewed.

1.2 Standard Error and the Central Limit Theorem

The standard deviation of a sampling distribution is called the **standard error (SE)**. It measures how much a statistic varies from sample to sample.

Sampling distribution of $\bar{x}$

- Mean: $\mu_{\bar{x}} = \mu$
- Standard error: $\sigma_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}}$
- Shape:
  - if the population is normal, $\bar{x}$ is normal for any n;
  - if not, $\bar{x}$ is **approximately normal** when n is large ($n \geq 30$). This is the **Central Limit Theorem (CLT)**.

So we can standardize $\bar{x}$ and use the z table:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Sampling distribution of $\hat{p}$

The sample proportion $\hat{p} = x/n$, where x is the number of "successes".

- Mean: $\mu_{\hat{p}} = p$
- Standard error: $\sigma_{\hat{p}} = \sqrt{\dfrac{p(1-p)}{n}}$
- Shape: approximately normal when $np \geq 5$ and $n(1-p) \geq 5$.

$$z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

Larger samples, smaller error

Because $\sqrt{n}$ is in the denominator, a sample four times as large halves the standard error.

Worked Example 1.1: daily wage

Suppose the daily wage of construction workers in Chiang Mai has mean $\mu = 380$ baht and standard deviation $\sigma = 60$ baht (hypothetical figures). A random sample of $n = 36$ workers is taken.

(a) Describe the sampling distribution of $\bar{x}$.
(b) Find the probability that the sample mean wage is more than 400 baht.

Solution

(a) Since $n = 36 \geq 30$, by the CLT $\bar{x}$ is approximately normal with

$$\mu_{\bar{x}} = 380, \qquad \sigma_{\bar{x}} = \frac{60}{\sqrt{36}} = 10 \text{ baht}.$$

(b)

$$z = \frac{400 - 380}{10} = 2.00$$

From Table 4, $P(Z \leq 2.00) = 0.9772$, so

$$P(\bar{x} > 400) = P(Z > 2.00) = 1 - 0.9772 = 0.0228.$$

About 2.3% of samples of 36 workers would have a mean wage above 400 baht.

Worked Example 1.2: household debt

Suppose 65% of households in a region have outstanding debt (hypothetical figure). A survey samples $n = 400$ households. Find the probability that less than 60% of the sampled households have debt.

Solution

Check the conditions: $np = 260 \geq 5$ and $n(1-p) = 140 \geq 5$, so $\hat{p}$ is approximately normal.

$$\sigma_{\hat{p}} = \sqrt{\frac{0.65(1 - 0.65)}{400}} = 0.0238$$

$$z = \frac{0.60 - 0.65}{0.0238} = -2.10$$

Table 4 gives only positive z. By symmetry,

$$P(\hat{p} < 0.60) = P(Z < -2.10) = P(Z > 2.10) = 1 - 0.9821 = 0.0179.$$

Your Turn

Your Turn 1.1

The monthly rent of rooms near a university has mean 4,500 baht and standard deviation 900 baht (hypothetical figures). A random sample of 81 rooms is taken. Find

(a) the standard error of $\bar{x}$, and
(b) the probability that the sample mean rent is less than 4,320 baht.

Your Turn 1.2

In a province, 40% of workers are in the informal sector (hypothetical figure). A labour force survey samples 600 workers. Find the probability that more than 45% of the sample work in the informal sector.

1.3 The Idea of a Confidence Interval

A single number such as $\bar{x}$ is a **point estimate**. It is almost never exactly equal to μ. A **confidence interval (CI)** gives a range of plausible values instead:

$$\text{point estimate} \pm (\text{critical value}) \times \text{SE}$$

The **confidence level** (for example 95%) describes the method, not one interval: if we took many samples and built a 95% CI from each, about 95% of the intervals would contain the true parameter.

In Figure 1.2, 37 of the 40 intervals contain $\mu = 380$ and 3 do not. For a single interval we write:

> We are 95% confident that the mean daily wage is between … and … baht.

> **A common mistake**
>
> Do not say "there is a 95% probability that μ is in this interval". Once the interval is calculated, μ is either in it or not. The 95% refers to how often the method works.

Chapters 2 and 3 give the formula of the CI for each parameter.

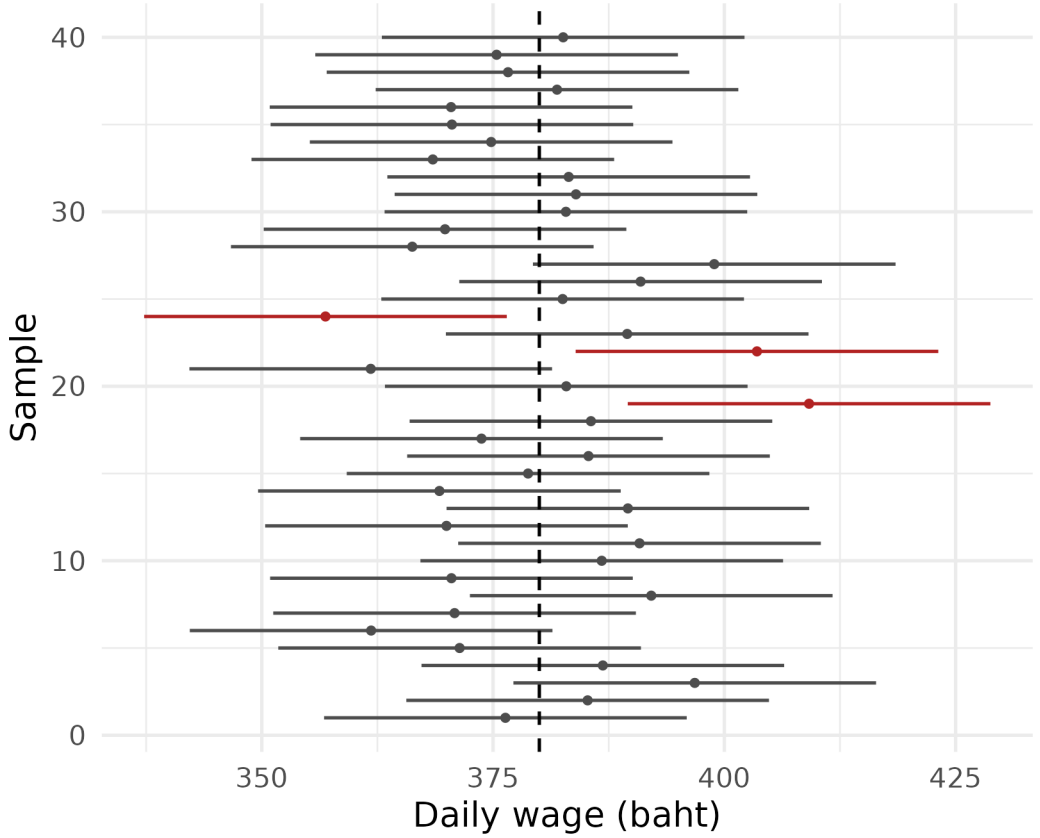


Figure 1.2: Forty 95% confidence intervals for the mean daily wage, each from a different sample of 36 workers (hypothetical population with mean 380 baht). Red intervals miss the true mean.

1.4 The Logic of Hypothesis Testing

A hypothesis test uses sample data to decide between two statements about a population parameter.

- The **null hypothesis** H_0 is the statement of "no change" or "no difference". It always contains the equals sign.
- The **alternative hypothesis** H_1 is what the researcher wants to show. It contains $\neq$, $<$ or $>$.

We assume H_0 is true and ask whether the sample result is unusual under that assumption. If it is, we reject H_0.

Example. The Ministry of Labour reports that the mean daily wage of construction workers is 380 baht. A labour union believes the mean is lower.

$$H_0 : \mu = 380 \qquad H_1 : \mu < 380$$

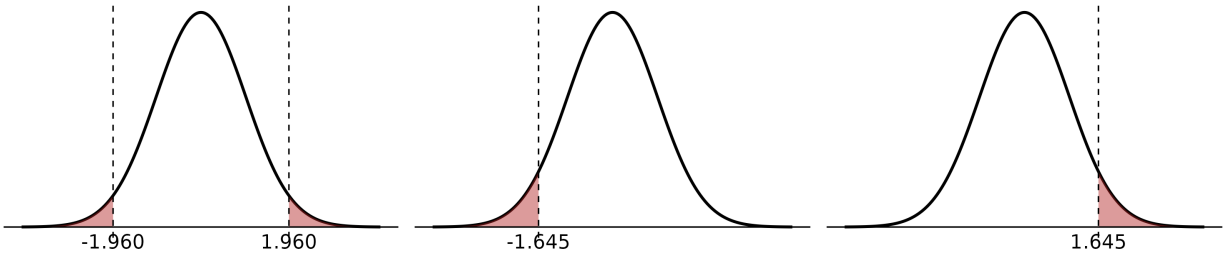
The direction in H_1 comes from the **research question**, never from the data.

Research question	H_1	Test	Critical region
Is it different from μ_0?	$\mu \neq \mu_0$	two-sided	both tails, area $\alpha/2$ each
Is it less than μ_0?	$\mu < \mu_0$	one-sided (left)	left tail, area α
Is it greater than μ_0?	$\mu > \mu_0$	one-sided (right)	right tail, area α

Test statistic, critical value and critical region

- The **test statistic** measures how far the sample result is from H_0, in standard errors. For example $z_{\text{cal}} = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$.
- The **level of significance** α is the probability of rejecting H_0 when H_0 is true. We usually use $\alpha = 0.05$.
- The **critical value** separates the values of the test statistic that lead to rejecting H_0 from those that do not.
- The **critical region** is the set of values that lead to rejecting H_0.

Decision rule. If z_{cal} falls in the critical region, we reject H_0. Otherwise we accept H_0.



(a) Two-sided (b) Left-sided (c) Right-sided

Figure 1.3: Critical regions for a z test at alpha = 0.05

Writing the hypotheses

Worked Example 1.3. Write H_0 and H_1.

(a) A price-monitoring agency checks whether the mean price of a 5-kg bag of rice has changed from 210 baht.
(b) An economist claims that more than 30% of households in Chiang Mai have used a government stimulus measure.
(c) A bank claims that the mean monthly loan payment of its customers is less than 8,000 baht.

Solution

(a) $H_0 : \mu = 210 \quad H_1 : \mu \neq 210$ (two-sided: "changed" has no direction)
(b) $H_0 : p = 0.30 \quad H_1 : p > 0.30$ (right-sided)
(c) $H_0 : \mu = 8000 \quad H_1 : \mu < 8000$ (left-sided)

Your Turn 1.3

Write H_0 and H_1, and say whether each test is two-sided, left-sided or right-sided.

(a) A researcher tests whether the mean household debt in a district is higher than 200,000 baht.
(b) An analyst tests whether the proportion of firms that raised prices last quarter is different from 0.50.
(c) A manager tests whether a new pricing strategy has reduced the mean number of days a product stays in stock below 14 days.

1.5 Type I and Type II Errors

The decision is based on a sample, so it can be wrong.

	H_0 is true	H_0 is false
Reject H_0	Type I error (probability α)	Correct decision
Accept H_0	Correct decision	Type II error (probability β)

Example. The government tests whether a stimulus measure increased mean household spending. H_0: the measure had no effect; H_1: the measure increased spending.

- **Type I error:** concluding that the measure worked when it did not. The government may continue an expensive policy that has no effect.
- **Type II error:** concluding that the measure did not work when it did. The government may stop a useful policy.

Choosing α

A smaller α makes a Type I error less likely but a Type II error more likely, for the same sample size. A larger sample reduces both.

1.6 Reading Statistical Tables

All the tables used in this course (Tables 4–7) give the **area to the left** of a value. The subscript in our notation is that area: $z_{0.975}$ is the value with area 0.975 to its left.

Two steps for every critical value

1. Draw the curve and **shade the area to the left** of the value you need.
2. Read the table at that left-tail area.

The z table (Table 4) gives $P(Z \leq z)$ for z from 0.00 to 3.49. The bottom row also lists the most common values:

Table 1.1: Common critical values of z (bottom row of Table 4)

Area to the left	z
0.900	1.282
0.950	1.645
0.975	1.960
0.990	2.326
0.995	2.576

Table 4 has only positive z. Because the normal curve is symmetric, $z_{0.05} = -z_{0.95}$ and $z_{0.025} = -z_{0.975}$.

Worked Example 1.4. Find the critical value(s) and critical region for a z test with $\alpha = 0.05$.

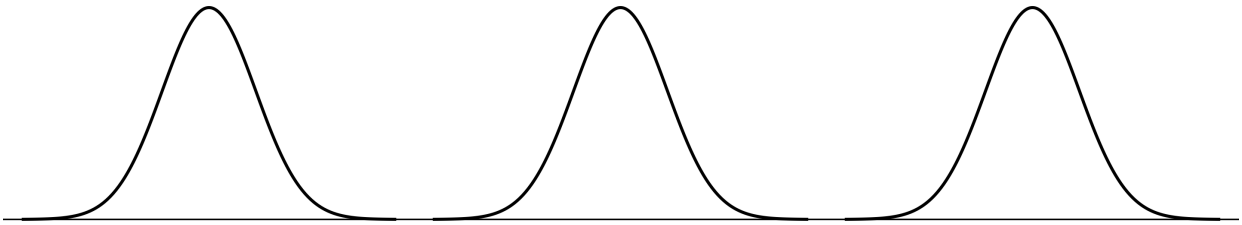
Solution

H_1	Area to the left	Critical value(s)	Critical region
$\neq$	0.025 and 0.975	$z_{0.025} = -1.960$, $z_{0.975} = 1.960$	$z_{\text{cal}} < -1.960$ or $z_{\text{cal}} > 1.960$
$<$	0.05	$z_{0.05} = -1.645$	$z_{\text{cal}} < -1.645$
$>$	0.95	$z_{0.95} = 1.645$	$z_{\text{cal}} > 1.645$

Your Turn 1.4

For each test, shade the area to the left on the curve first, then find the critical value(s) and write the critical region.

(a) $H_1 : \mu \neq \mu_0$, $\alpha = 0.01$
(b) $H_1 : \mu > \mu_0$, $\alpha = 0.01$
(c) $H_1 : p < p_0$, $\alpha = 0.10$



(a) (b) (c)

Figure 1.4: Shade the area to the left before reading Table 4

Formula Summary

Statistic	Mean	Standard error	Normal when	z
$\bar{x}$	μ	$\frac{\sigma}{\sqrt{n}}$	population normal, or $n \geq 30$	$\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$
$\hat{p}$	p	$\sqrt{\frac{p(1-p)}{n}}$	$np \geq 5$ and $n(1-p) \geq 5$	$\frac{\hat{p}-p}{\sqrt{p(1-p)/n}}$

H_1	Critical region (z test)
$\neq$	$z_{\text{cal}} < z_{\alpha/2}$ or $z_{\text{cal}} > z_{1-\alpha/2}$
$<$	$z_{\text{cal}} < z_{\alpha}$
$>$	$z_{\text{cal}} > z_{1-\alpha}$

Exercises

1.1 The price of a kilogram of pork at fresh markets has mean 45 baht and standard deviation 4 baht (hypothetical figures). A random sample of 64 markets is taken.

(a) Find the standard error of $\bar{x}$.
(b) Find $P(\bar{x} < 44.2)$.
(c) Find $P(44.2 < \bar{x} < 46)$.

1.2 Write H_0 and H_1.

(a) The Bank of Thailand tests whether mean headline inflation over the past year was above its 3% upper target.
(b) A researcher tests whether the proportion of graduates who find a job within six months differs from 0.70.
(c) A firm tests whether a discount campaign reduced the mean time to sell a product below 20 days.

1.3 Find the critical value(s) and the critical region of a z test.

(a) $H_1 : \mu \neq \mu_0$, $\alpha = 0.10$
(b) $H_1 : \mu < \mu_0$, $\alpha = 0.01$
(c) $H_1 : p > p_0$, $\alpha = 0.05$

1.4 A provincial office tests whether a job-training programme raised the mean monthly income of participants. H_0: the programme has no effect on mean income; H_1: the programme raises mean income. Describe a Type I error and a Type II error in this context, and give one consequence of each.

1.5 The unemployment rate in a province is 5% (hypothetical figure). Can the normal approximation be used for $\hat{p}$ when (a) $n = 80$, (b) $n = 200$? Explain.

1.6 A 95% CI for the mean monthly electricity bill of households in a city is 1,120 to 1,280 baht. Which statements are correct?

(a) We are 95% confident that the mean monthly electricity bill of all households in the city is between 1,120 and 1,280 baht.
(b) 95% of households in the city pay between 1,120 and 1,280 baht per month.
(c) If we took many samples and built a 95% CI from each, about 95% of the intervals would contain the population mean.

Solutions

Solution: Your Turn 1.1

(a) $\sigma_{\bar{x}} = 900/\sqrt{81} = 100$ baht.

(b) $z = (4320 - 4500)/100 = -1.80$, so $P(\bar{x} < 4320) = P(Z > 1.80) = 1 - 0.9641 = 0.0359$.

Solution: Your Turn 1.2

$np = 240$ and $n(1-p) = 360$ are both at least 5. $\sigma_{\hat{p}} = \sqrt{0.4(0.6)/600} = 0.0200$, $z = (0.45 - 0.4)/0.0200 = 2.50$, so $P(\hat{p} > 0.45) = 1 - 0.9938 = 0.0062$.

Solution: Your Turn 1.3

(a) $H_0 : \mu = 200000 \quad H_1 : \mu > 200000$ (right-sided)
(b) $H_0 : p = 0.50 \quad H_1 : p \neq 0.50$ (two-sided)
(c) $H_0 : \mu = 14 \quad H_1 : \mu < 14$ (left-sided)

Solution: Your Turn 1.4

(a) Areas 0.005 and 0.995: $z_{0.995} = 2.576$. Critical region: $z_{\text{cal}} < -2.576$ or $z_{\text{cal}} > 2.576$.

(b) Area 0.99: $z_{0.99} = 2.326$. Critical region: $z_{\text{cal}} > 2.326$.

(c) Area 0.10: $z_{0.10} = -z_{0.90} = -1.282$. Critical region: $z_{\text{cal}} < -1.282$.

Solution: Exercise 1.1

(a) $\sigma_{\bar{x}} = 4/\sqrt{64} = 0.5$ baht.

(b) $z = (44.2 - 45)/0.5 = -1.60$; $P(\bar{x} < 44.2) = 1 - 0.9452 = 0.0548$.

(c) $z = (46 - 45)/0.5 = 2.00$; $P(44.2 < \bar{x} < 46) = 0.9772 - 0.0548 = 0.9224$.

Solution: Exercise 1.2

(a) $H_0 : \mu = 3 \quad H_1 : \mu > 3$
(b) $H_0 : p = 0.70 \quad H_1 : p \neq 0.70$
(c) $H_0 : \mu = 20 \quad H_1 : \mu < 20$

Solution: Exercise 1.3

(a) $z_{0.95} = 1.645$: reject H_0 if $z_{\text{cal}} < -1.645$ or $z_{\text{cal}} > 1.645$.
(b) $z_{0.01} = -z_{0.99} = -2.326$: reject H_0 if $z_{\text{cal}} < -2.326$.
(c) $z_{0.95} = 1.645$: reject H_0 if $z_{\text{cal}} > 1.645$.

Solution: Exercise 1.4

Type I error: concluding that the programme raises income when it does not. Budget may continue to be spent on a programme that does not work. Type II error: concluding that the programme has no effect when it does raise income. A useful programme may be cut.

Solution: Exercise 1.5

(a) $np = 80(0.05) = 4 < 5$, so the normal approximation should not be used.
(b) $np = 200(0.05) = 10$ and $n(1-p) = 190$ are both at least 5, so it can be used.

Solution: Exercise 1.6

(a) and (c) are correct. (b) is wrong: the interval is about the population mean, not about individual households.

2 One-Sample Inference

2.1 Mean, σ Known

Learning Objectives

1. Recognize when the z method for one mean applies.
2. Construct and interpret a confidence interval for μ when σ is known.
3. Carry out a z test for μ in five steps and write the conclusion.

Concept

Sometimes the population standard deviation σ is known from a large source, such as a census or many years of administrative records, while the mean may have changed. For example, the spread of daily wages in an industry is fairly stable, but the mean wage changes with prices and minimum-wage policy.

When σ is known, Chapter 1 tells us that

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

follows the standard normal distribution, provided the population is normal or $n \geq 30$. Both the confidence interval and the test use this result.

Formulas and Conditions

Conditions: σ is known; a random sample; the population is normal or $n \geq 30$.

$(1-\alpha)100\%$ **confidence interval for** μ

$$\bar{x} \pm z_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}}$$

For a 95% CI, $z_{0.975} = 1.960$; for a 99% CI, $z_{0.995} = 2.576$.

Test statistic for $H_0 : \mu = \mu_0$

$$z_{\text{cal}} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

H_1	Critical region
$\mu \neq \mu_0$	$z_{\text{cal}} < z_{\alpha/2}$ or $z_{\text{cal}} > z_{1-\alpha/2}$
$\mu < \mu_0$	$z_{\text{cal}} < z_{\alpha}$
$\mu > \mu_0$	$z_{\text{cal}} > z_{1-\alpha}$

Worked Example

Worked Example 2.1.1. From many years of Labour Force Survey records, the standard deviation of the daily wage of construction workers in Chiang Mai is $\sigma = 60$ baht. A labour union surveys 36 workers and finds a mean daily wage of $\bar{x} = 358.4$ baht (hypothetical figures). The Ministry of Labour reports a mean of 380 baht; the union believes the true mean is lower.

(a) Construct a 95% confidence interval for the mean daily wage.
(b) At $\alpha = 0.05$, test whether the mean daily wage is less than 380 baht.

Solution

(a) $n = 36 \geq 30$, so the CLT applies.

$$\frac{\sigma}{\sqrt{n}} = \frac{60}{\sqrt{36}} = 10$$

$$\bar{x} \pm z_{0.975}\frac{\sigma}{\sqrt{n}} = 358.4 \pm 1.960(10) = 358.4 \pm 19.6$$

95% CI: (338.80, 378.00)
We are 95% confident that the mean daily wage of construction workers in Chiang Mai is between 338.80 and 378.00 baht.
(b)
Step 1: Hypotheses. $H_0 : \mu = 380 \quad H_1 : \mu < 380$
Step 2: Level of significance. $\alpha = 0.05$
Step 3: Test statistic.

$$z_{\text{cal}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{358.4 - 380}{10} = -2.160$$

Step 4: Critical value and critical region. Left-sided test, area 0.05 to the left: $z_{0.05} = -z_{0.95} = -1.645$. Critical region: $z_{\text{cal}} < -1.645$.

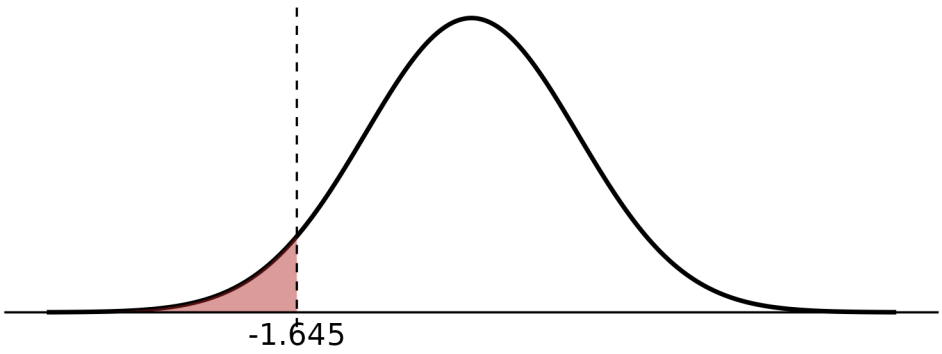


Figure 2.1: Critical region for Worked Example 2.1.1

Step 5: Decision and interpretation. $z_{\text{cal}} = -2.160$ falls in the critical region ($-2.160 < -1.645$), so we reject H_0. That is, the mean daily wage of construction workers in Chiang Mai is less than 380 baht at the 0.05 level of significance.

Your Turn

Your Turn 2.1.1

The standard deviation of the price of a 5-kg bag of jasmine rice across shops is known to be $\sigma = 12$ baht. A price-monitoring agency checks 49 shops and finds a mean price of $\bar{x} = 214.6$ baht (hypothetical figures).

(a) Construct a 95% confidence interval for the mean price.
(b) At $\alpha = 0.05$, test whether the mean price is different from 210 baht.

Step 1: Hypotheses

H_0: _____

H_1: _____

Step 2: Level of significance

$\alpha =$ _____

Step 3: Test statistic

Formula: _____

Substitution: _____

Result: _____

Step 4: Critical value and critical region

Critical value(s): _____

Critical region: _____

Step 5: Decision and interpretation

Decision: _____

_____

Interpretation: _____

_____

Formula Summary

	Formula
Conditions	σ known; population normal or $n \geq 30$
$(1-\alpha)100\%$ CI for μ	$\bar{x} \pm z_{1-\alpha/2}\,\sigma/\sqrt{n}$
Test statistic	$z_{\text{cal}} = \dfrac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$
Critical region	$\neq$: $z_{\text{cal}} < z_{\alpha/2}$ or $z_{\text{cal}} > z_{1-\alpha/2}$; $<$: $z_{\text{cal}} < z_{\alpha}$; $>$: $z_{\text{cal}} > z_{1-\alpha}$

Exercises

2.1.1 Utility records show that the standard deviation of the monthly electricity bill of households in a city is 320 baht. A sample of 64 households has a mean bill of 1,240 baht (hypothetical figures). Construct a 99% confidence interval for the mean monthly bill.

2.1.2 A company claims that the mean monthly sales per salesperson are more than 150 thousand baht. The standard deviation is known to be 24 thousand baht. A sample of 40 salespeople has a mean of 156.2 thousand baht (hypothetical figures). Test the claim at $\alpha = 0.01$.

2.1.3 The standard deviation of savings deposits at a bank is known to be 12,000 baht. A sample of 100 accounts has a mean balance of 52,100 baht (hypothetical figures).

(a) At $\alpha = 0.05$, test whether the mean balance is different from 50,000 baht.
(b) Construct a 95% CI for the mean balance. Does it agree with (a)?

Solutions

Solution: Your Turn 2.1.1

(a) $\sigma/\sqrt{n} = 12/\sqrt{49} = 1.7143$; $214.6 \pm 1.960(1.7143) = 214.6 \pm 3.36$. 95% CI: (211.24, 217.96). We are 95% confident that the mean price of a 5-kg bag of jasmine rice is between 211.24 and 217.96 baht.

(b)

Step 1. $H_0 : \mu = 210 \quad H_1 : \mu \neq 210$

Step 2. $\alpha = 0.05$

Step 3. $z_{\text{cal}} = \dfrac{214.6 - 210}{1.7143} = 2.683$

Step 4. $z_{0.025} = -1.960$, $z_{0.975} = 1.960$. Critical region: $z_{\text{cal}} < -1.960$ or $z_{\text{cal}} > 1.960$.

Step 5. $z_{\text{cal}} = 2.683$ falls in the critical region ($2.683 > 1.960$), so we reject H_0. That is, the mean price of a 5-kg bag of jasmine rice is different from 210 baht at the 0.05 level of significance; since $\bar{x} = 214.6$, the mean price is higher than 210 baht.

Solution: Exercise 2.1.1

$\sigma/\sqrt{n} = 320/\sqrt{64} = 40$; $z_{0.995} = 2.576$. $1240 \pm 2.576(40) = 1240 \pm 103.04$. 99% CI: (1,136.96, 1,343.04). We are 99% confident that the mean monthly electricity bill is between 1,136.96 and 1,343.04 baht.

Solution: Exercise 2.1.2

Step 1. $H_0 : \mu = 150 \quad H_1 : \mu > 150$

Step 2. $\alpha = 0.01$

Step 3. $\sigma/\sqrt{n} = 24/\sqrt{40} = 3.7947$; $z_{\text{cal}} = (156.2 - 150)/3.7947 = 1.634$

Step 4. $z_{0.99} = 2.326$. Critical region: $z_{\text{cal}} > 2.326$.

Step 5. $z_{\text{cal}} = 1.634$ does not fall in the critical region ($1.634 < 2.326$), so we accept H_0. That is, the mean monthly sales per salesperson are equal to 150 thousand baht at the 0.01 level of significance.

Solution: Exercise 2.1.3

(a) $H_0 : \mu = 5 \times 10^4 \quad H_1 : \mu \neq 5 \times 10^4$; $\alpha = 0.05$. $\sigma/\sqrt{n} = 1200$; $z_{\text{cal}} = (5.21 \times 10^4 - 5 \times 10^4)/1200 = 1.750$. Critical region: $z_{\text{cal}} < -1.960$ or $z_{\text{cal}} > 1.960$. z_{cal} does not fall in the critical region ($-1.960 < 1.750 < 1.960$), so we accept H_0. That is, the mean balance is equal to 50,000 baht at the 0.05 level of significance.

(b) $5.21 \times 10^4 \pm 1.960(1200)$: 95% CI (49,748.00, 54,452.00). The interval contains 50,000, which agrees with accepting H_0 in (a).

2.2 Mean, σ Unknown

Learning Objectives

1. Use the t distribution when σ is unknown and read the t table (Table 5).
2. Construct and interpret a confidence interval for μ using s.
3. Carry out a t test for μ in five steps.
4. Use a confidence interval to reach the decision of a two-sided test.

Concept

In practice σ is almost never known. We replace it by the sample standard deviation s. Because s itself varies from sample to sample, the statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

is more spread out than z. It follows a t **distribution** with $\text{df} = n - 1$ degrees of freedom, provided the population is normal. For moderate or large n the method still works well when the population is not exactly normal.

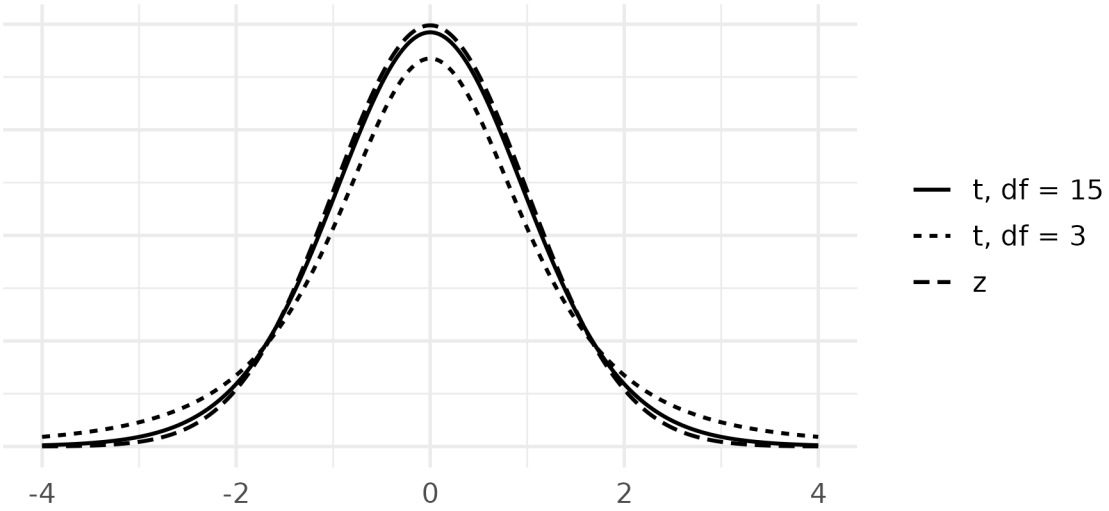


Figure 2.2: The t distribution has heavier tails than z; with more df it gets closer to z

Formulas and Conditions

Conditions: σ unknown; a random sample; the population is approximately normal (less important when n is large).

$(1-\alpha)100\%$ **confidence interval for** μ

$$\bar{x} \pm t_{1-\alpha/2,\, n-1}\, \frac{s}{\sqrt{n}}$$

Test statistic for $H_0 : \mu = \mu_0$, with $\text{df} = n - 1$

$$t_{\text{cal}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

H_1	Critical region
$\mu \neq \mu_0$	$t_{\text{cal}} < t_{\alpha/2,\, n-1}$ or $t_{\text{cal}} > t_{1-\alpha/2,\, n-1}$
$\mu < \mu_0$	$t_{\text{cal}} < t_{\alpha,\, n-1}$
$\mu > \mu_0$	$t_{\text{cal}} > t_{1-\alpha,\, n-1}$

Reading the t Table

Table 5 gives t values with a stated **area to the left** (columns) for each df (rows). Like z, the t curve is symmetric, so the table lists only the right half: $t_{0.05,\,\text{df}} = -t_{0.95,\,\text{df}}$.

Not every df is in the table

After df = 30 the table jumps to 40, 50, 60, 80, 100, 200, 500 and ∞. All examples in this book use df that appear in the table.

Worked Example 2.2.1. Find (a) $t_{0.975,\,24}$, (b) $t_{0.05,\,15}$, (c) $t_{0.995,\,40}$.

Solution

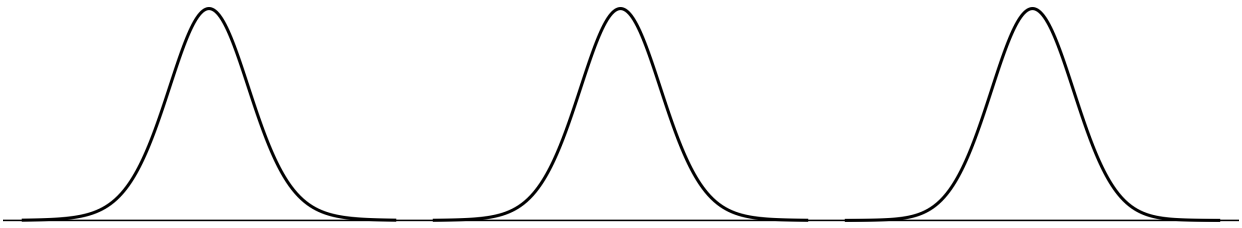
(a) Row df = 24, column 0.975: $t_{0.975,\,24} = 2.064$.

(b) Area 0.05 is in the left tail. By symmetry, $t_{0.05,\,15} = -t_{0.95,\,15} = -1.753$.

(c) Row df = 40, column 0.995: $t_{0.995,\,40} = 2.704$.

Your Turn 2.2.1

Shade the area to the left on each curve, then find (a) $t_{0.975,\,15}$, (b) $t_{0.01,\,20}$, (c) $t_{0.95,\,60}$.



(a) (b) (c)

Figure 2.3: Shade the area to the left before reading Table 5

Worked Example

Worked Example 2.2.2. A food-delivery platform claims that its riders in Chiang Mai earn more than 600 baht per day on average. A researcher records the daily earnings (baht) of 25 randomly selected riders (hypothetical data):

Table 2.1: Daily earnings (baht) of 25 riders

724	756	588	693	777
657	631	777	549	444
511	614	754	593	698
541	616	793	670	698
555	668	510	740	567

The sample gives $\bar{x} = 644.96$ and $s = 96.6499$ baht.

(a) Construct a 95% confidence interval for the mean daily earnings.
(b) At $\alpha = 0.05$, test the platform's claim.

Solution

(a) $\text{df} = 25 - 1 = 24$ and $t_{0.975,\,24} = 2.064$.

$$\frac{s}{\sqrt{n}} = \frac{96.6499}{\sqrt{25}} = 19.33$$

$$\bar{x} \pm t_{0.975,\,24}\frac{s}{\sqrt{n}} = 644.96 \pm 2.064(19.33) = 644.96 \pm 39.8971$$

95% CI: (605.06, 684.86)
We are 95% confident that the mean daily earnings of riders in Chiang Mai are between 605.06 and 684.86 baht.
(b)
Step 1: Hypotheses. $H_0 : \mu = 600 \quad H_1 : \mu > 600$
Step 2: Level of significance. $\alpha = 0.05$
Step 3: Test statistic.

$$t_{\text{cal}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{644.96 - 600}{19.33} = 2.326$$

Step 4: Critical value and critical region. Right-sided test with $\text{df} = 24$: $t_{0.95,\,24} = 1.711$. Critical region: $t_{\text{cal}} > 1.711$.

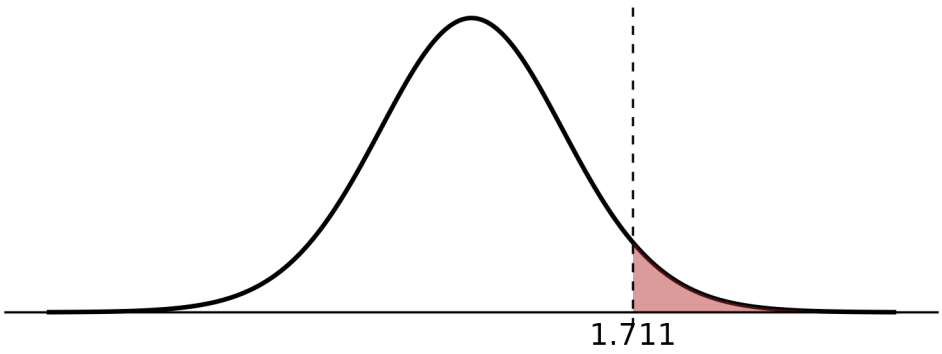


Figure 2.4: Critical region for Worked Example 2.2.2

Step 5: Decision and interpretation. $t_{\text{cal}} = 2.326$ falls in the critical region ($2.326 > 1.711$), so we reject H_0. That is, the mean daily earnings of riders in Chiang Mai are more than 600 baht at the 0.05 level of significance.

Confidence Intervals and Two-Sided Tests

A $(1-\alpha)100\%$ CI and a **two-sided** test at level α always agree:

- if μ_0 is **outside** the CI, a two-sided test rejects $H_0 : \mu = \mu_0$;
- if μ_0 is **inside** the CI, the test accepts H_0.

For the riders, the 95% CI (605.06, 684.86) does not contain 600, so a two-sided test of $H_0 : \mu = 600$ at $\alpha = 0.05$ would reject H_0. It does contain 620, so the same test of $H_0 : \mu = 620$ would accept H_0.

Only for two-sided tests

This shortcut matches a two-sided test with the same α as the confidence level ($\alpha = 0.05$ with a 95% CI). For a one-sided test, carry out the five steps.

Your Turn

Your Turn 2.2.2

A property agent records the monthly rent of 16 one-bedroom condominium units near Nimmanhaemin Road, Chiang Mai. The sample mean is 9,420 baht and the standard deviation is 784 baht (hypothetical figures). Assume rents are normally distributed.

(a) Construct a 95% confidence interval for the mean rent.
(b) At $\alpha = 0.05$, test whether the mean rent is different from 9,000 baht.

(c) Does the CI in (a) agree with your decision in (b)?

Step 1: Hypotheses

H_0:

H_1:

Step 2: Level of significance

$\alpha =$

Step 3: Test statistic

Formula:

Substitution:

Result:

Step 4: Critical value and critical region

Critical value(s): _____

Critical region: _____

Step 5: Decision and interpretation

Decision: _____

_____

Interpretation: _____

_____

Formula Summary

	Formula
Conditions	σ unknown; population approximately normal
$(1-\alpha)100\%$ CI for μ	$\bar{x} \pm t_{1-\alpha/2,\,n-1}\, s/\sqrt{n}$
Test statistic	$t_{\text{cal}} = \dfrac{\bar{x}-\mu_0}{s/\sqrt{n}}$, $\text{df} = n-1$
Critical region	$\neq$: $t_{\text{cal}} < t_{\alpha/2,\,n-1}$ or $t_{\text{cal}} > t_{1-\alpha/2,\,n-1}$; $<$: $t_{\text{cal}} < t_{\alpha,\,n-1}$; $>$: $t_{\text{cal}} > t_{1-\alpha,\,n-1}$
CI and test	two-sided test at α rejects $H_0 \iff \mu_0$ is outside the $(1-\alpha)100\%$ CI

Exercises

2.2.1 A survey of 20 households in a district finds a mean household debt of 184,500 baht with a standard deviation of 41,200 baht (hypothetical figures). At $\alpha = 0.05$, test whether the mean household debt is more than 170,000 baht.

2.2.2 A sample of 30 gig workers reports a mean of 47.3 working hours per week with a standard deviation of 8.4 hours (hypothetical figures). Construct a 99% confidence interval for the mean weekly working hours.

2.2.3 A 95% confidence interval for the mean monthly interest rate charged by informal lenders is (12.4%, 15.8%) (hypothetical figures). Without further calculation, what is the decision of a two-sided test at $\alpha = 0.05$ for (a) $H_0 : \mu = 16$, (b) $H_0 : \mu = 15$?

2.2.4 Find (a) $t_{0.90,\,12}$, (b) $t_{0.025,\,8}$, (c) $t_{0.99,\,100}$.

Solutions

Solution: Your Turn 2.2.1

(a) $t_{0.975,\,15} = 2.131$

(b) $t_{0.01,\,20} = -t_{0.99,\,20} = -2.528$

(c) $t_{0.95,\,60} = 1.671$

Solution: Your Turn 2.2.2

(a) df $= 15$, $t_{0.975,\,15} = 2.131$; $s/\sqrt{n} = 784/\sqrt{16} = 196$; $9420 \pm 2.131(196) = 9420 \pm 417.676$. 95% CI: (9,002.32, 9,837.68). We are 95% confident that the mean monthly rent is between 9,002.32 and 9,837.68 baht.

(b)

Step 1. $H_0 : \mu = 9000 \quad H_1 : \mu \neq 9000$

Step 2. $\alpha = 0.05$

Step 3. $t_{\text{cal}} = \dfrac{9420 - 9000}{196} = 2.143$

Step 4. df $= 15$: $t_{0.025,\,15} = -2.131$, $t_{0.975,\,15} = 2.131$. Critical region: $t_{\text{cal}} < -2.131$ or $t_{\text{cal}} > 2.131$.

Step 5. $t_{\text{cal}} = 2.143$ falls in the critical region ($2.143 > 2.131$), so we reject H_0. That is, the mean monthly rent is different from 9,000 baht at the 0.05 level of significance; since $\bar{x}$ = 9,420, the mean rent is higher than 9,000 baht.

(c) Yes. The 95% CI does not contain 9,000, so the two-sided test at $\alpha = 0.05$ rejects H_0.

Solution: Exercise 2.2.1

Step 1. $H_0 : \mu = 1.7 \times 10^5 \quad H_1 : \mu > 1.7 \times 10^5$

Step 2. $\alpha = 0.05$

Step 3. $s/\sqrt{n} = 4.12 \times 10^4/\sqrt{20} = 9212.6001$; $t_{\text{cal}} = (1.845 \times 10^5 - 1.7 \times 10^5)/9212.6001 = 1.574$

Step 4. df $= 19$: $t_{0.95,\,19} = 1.729$. Critical region: $t_{\text{cal}} > 1.729$.

Step 5. $t_{\text{cal}} = 1.574$ does not fall in the critical region ($1.574 < 1.729$), so we accept H_0. That is, the mean household debt in the district is equal to 170,000 baht at the 0.05 level of significance.

Solution: Exercise 2.2.2

df $= 29$, $t_{0.995,\,29} = 2.756$; $s/\sqrt{n} = 8.4/\sqrt{30} = 1.5336$; $47.3 \pm 2.756(1.5336) = 47.3 \pm 4.2266$. 99% CI: (43.07, 51.53). We are 99% confident that the mean weekly working hours of gig workers are between 43.07 and 51.53 hours.

Solution: Exercise 2.2.3

(a) 16 is outside the CI, so we reject H_0.
(b) 15 is inside the CI, so we accept H_0.

Solution: Exercise 2.2.4

(a) 1.356 (b) $-t_{0.975,\,8} = -2.306$ (c) 2.364

2.3 Proportion

Learning Objectives

1. Check the conditions for the normal approximation to $\hat{p}$.
2. Construct and interpret a confidence interval for a population proportion p.
3. Carry out a z test for p in five steps.
4. Explain why the CI uses $\hat{p}$ in the standard error but the test uses p_0.

Concept

Many economic questions are about proportions: the share of households in debt, the labour force participation rate, the share of firms that raised prices, or the share of people who used a stimulus measure. Each person or firm either has the characteristic ("success") or not.

With x successes in a sample of n, the sample proportion is $\hat{p} = x/n$. From Chapter 1, $\hat{p}$ is approximately normal with mean p and standard error $\sqrt{p(1-p)/n}$ when the sample is large enough.

Which proportion goes in the standard error?

The standard error contains the unknown p, so we must put in a value:

- **Confidence interval:** we do not assume any value of p, so we use our best estimate $\hat{p}$.
- **Hypothesis test:** we assume H_0 is true, so we use p_0.

Formulas and Conditions

$(1-\alpha)100\%$ **confidence interval for** p

$$\hat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \qquad \text{condition: } n\hat{p} \geq 5 \text{ and } n(1-\hat{p}) \geq 5$$

Test statistic for $H_0 : p = p_0$

$$z_{\text{cal}} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \qquad \text{condition: } np_0 \geq 5 \text{ and } n(1-p_0) \geq 5$$

The critical regions are the same as for the z test of a mean.

Worked Example

Worked Example 2.3.1. The government ran a co-payment scheme in which it paid part of the price of goods bought at small shops. An economist claims that more than 30% of households in Chiang Mai used the scheme. In a random sample of 400 households, 148 used it (hypothetical figures).

(a) Construct a 95% confidence interval for the proportion of households that used the scheme.
(b) At $\alpha = 0.05$, test the economist's claim.

Solution

$\hat{p} = 148/400 = 0.37$
(a) Check: $n\hat{p} = 148 \geq 5$ and $n(1-\hat{p}) = 252 \geq 5$.

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.37(0.63)}{400}} = 0.0241$$

$$0.37 \pm 1.960(0.0241) = 0.37 \pm 0.0472$$

95% CI: (0.3228, 0.4172)
We are 95% confident that the proportion of households in Chiang Mai that used the scheme is between 0.3228 and 0.4172.
(b)
Step 1: Hypotheses. $H_0 : p = 0.30 \quad H_1 : p > 0.30$
Step 2: Level of significance. $\alpha = 0.05$
Step 3: Test statistic. Check: $np_0 = 120 \geq 5$ and $n(1-p_0) = 280 \geq 5$.

$$\sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.30(0.70)}{400}} = 0.0229$$

$$z_{\text{cal}} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} = \frac{0.37 - 0.30}{0.0229} = 3.057$$

Step 4: Critical value and critical region. $z_{0.95} = 1.645$. Critical region: $z_{\text{cal}} > 1.645$.

Step 5: Decision and interpretation. $z_{\text{cal}} = 3.057$ falls in the critical region ($3.057 > 1.645$), so we reject H_0. That is, the proportion of households in Chiang Mai that used the scheme is more than 0.30 at the 0.05 level of significance.

Your Turn

Your Turn 2.3.1

A university reports that 70% of its economics graduates find a job within six months. A survey of 250 recent graduates finds that 160 found a job within six months (hypothetical figures).

(a) Construct a 95% confidence interval for the proportion of graduates who find a job within six months.
(b) At $\alpha = 0.05$, test whether the proportion is different from 0.70.

Step 1: Hypotheses

Step 2: Level of significance

Step 3: Test statistic

Step 4: Critical value and critical region

Step 5: Decision and interpretation

Formula Summary

	Confidence interval	Hypothesis test
Standard error	$\sqrt{\hat{p}(1-\hat{p})/n}$	$\sqrt{p_0(1-p_0)/n}$
Condition	$n\hat{p} \geq 5$, $n(1-\hat{p}) \geq 5$	$np_0 \geq 5$, $n(1-p_0) \geq 5$
Formula	$\hat{p} \pm z_{1-\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$	$z_{\text{cal}} = \dfrac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$

Exercises

2.3.1 In a survey of 500 small businesses, 95 had borrowed from an informal lender in the past year (hypothetical figures). Construct a 99% confidence interval for the proportion of small businesses that borrowed from an informal lender.

2.3.2 A bank believes that less than 25% of its credit-card holders pay only the minimum amount each month. In a sample of 300 card holders, 66 paid only the minimum (hypothetical figures). Test the bank's belief at $\alpha = 0.05$.

2.3.3 A researcher wants to test $H_0 : p = 0.05$ for the unemployment rate in a district using a sample of 60 people. Is the z test appropriate? Explain.

Solutions

Solution: Your Turn 2.3.1

$\hat{p} = 160/250 = 0.64$

(a) $\sqrt{0.64(0.36)/250} = 0.0304$; $0.64 \pm 1.960(0.0304) = 0.64 \pm 0.0596$. 95% CI: (0.5804, 0.6996). We are 95% confident that the proportion of graduates who find a job within six months is between 0.5804 and 0.6996.

(b)

Step 1. $H_0 : p = 0.70 \quad H_1 : p \neq 0.70$

Step 2. $\alpha = 0.05$

Step 3. $\sqrt{0.70(0.30)/250} = 0.029$; $z_{\text{cal}} = (0.64 - 0.70)/0.029 = -2.069$

Step 4. Critical region: $z_{\text{cal}} < -1.960$ or $z_{\text{cal}} > 1.960$.

Step 5. $z_{\text{cal}} = -2.069$ falls in the critical region ($-2.069 < -1.960$), so we reject H_0. That is, the proportion of graduates who find a job within six months is different from 0.70 at the 0.05 level of significance; since $\hat{p} = 0.64$, the proportion is lower than 0.70.

Solution: Exercise 2.3.1

$\hat{p} = 0.19$; $\sqrt{0.19(0.81)/500} = 0.0175$; $z_{0.995} = 2.576$; $0.19 \pm 2.576(0.0175) = 0.19 \pm 0.0451$. 99% CI: (0.1449, 0.2351). We are 99% confident that the proportion of small businesses that borrowed from an informal lender is between 0.1449 and 0.2351.

Solution: Exercise 2.3.2

Step 1. $H_0 : p = 0.25 \quad H_1 : p < 0.25$

Step 2. $\alpha = 0.05$

Step 3. $\hat{p} = 0.22$; $\sqrt{0.25(0.75)/300} = 0.025$; $z_{\text{cal}} = (0.22 - 0.25)/0.025 = -1.200$

Step 4. $z_{0.05} = -1.645$. Critical region: $z_{\text{cal}} < -1.645$.

Step 5. $z_{\text{cal}} = -1.200$ does not fall in the critical region ($-1.200 > -1.645$), so we accept H_0. That is, the proportion of card holders who pay only the minimum is equal to 0.25 at the 0.05 level of significance.

Solution: Exercise 2.3.3

No. $np_0 = 60(0.05) = 3 < 5$, so the normal approximation is not reliable. A larger sample (at least 100 people) is needed.

2.4 Variance

Learning Objectives

1. Explain why the variance matters in economics (risk, volatility, price stability).
2. Read the chi-square table (Table 6).
3. Construct and interpret a confidence interval for σ^2 and σ.
4. Carry out a chi-square test for σ^2 in five steps.

Concept

The mean tells us the typical value; the variance tells us how much values **vary**. In finance the standard deviation of returns is the most common measure of **risk** (volatility). A central bank cares about the stability of prices, and a firm cares about the stability of its sales.

To make inferences about σ^2 we use the sample variance s^2. If the population is normal,

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

follows a **chi-square distribution** with $n-1$ degrees of freedom. The chi-square distribution takes only positive values and is skewed to the right.

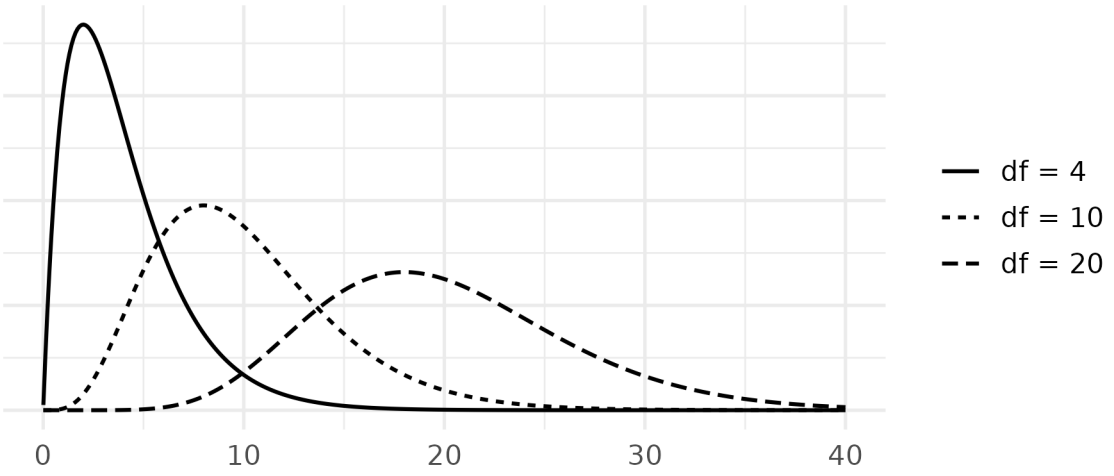


Figure 2.5: Chi-square distributions are skewed to the right; the skew decreases as df increases

> The population must be normal
>
> The chi-square methods for a variance are sensitive to non-normality, even for large samples. Use them only when the data are approximately normal.

Reading the Chi-Square Table

Table 6 gives χ^2 values with a stated area to the left, for df $= 1$ to 100. The curve is **not symmetric**, so for a two-sided test or a CI we read two different values from the table; we cannot use a minus sign.

Worked Example 2.4.1. For df $= 23$, find $\chi^2_{0.025,\,23}$, $\chi^2_{0.975,\,23}$ and $\chi^2_{0.95,\,23}$.

> Solution
>
> Row df $= 23$: $\chi^2_{0.025,\,23} = 11.7$, $\chi^2_{0.975,\,23} = 38.1$ and $\chi^2_{0.95,\,23} = 35.2$.

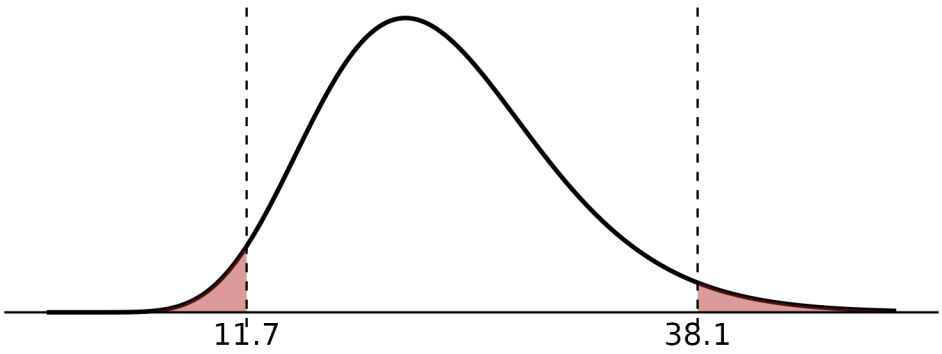
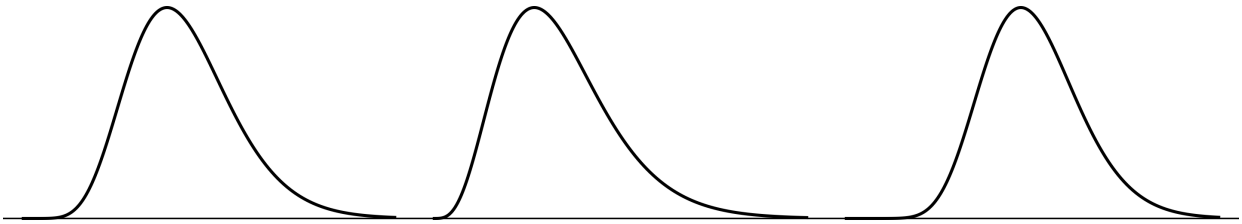


Figure 2.6: Areas 0.025 in each tail of the chi-square distribution with df = 23

Your Turn 2.4.1

Shade the area to the left on each curve, then find (a) $\chi^2_{0.05,\,19}$, (b) $\chi^2_{0.995,\,10}$, (c) $\chi^2_{0.025,\,30}$.



(a) (b) (c)

Figure 2.7: Shade the area to the left before reading Table 6

Formulas and Conditions

Condition: a random sample from a normal population.

$(1-\alpha)100\%$ **confidence interval for** σ^2

$$\left(\frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}},\ \frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}}\right)$$

Take the square root of both limits to get a CI for σ. The larger chi-square value gives the **lower** limit. The interval is not symmetric about s^2.

Test statistic for $H_0 : \sigma^2 = \sigma_0^2$, with df $= n-1$

$$\chi^2_{\text{cal}} = \frac{(n-1)s^2}{\sigma_0^2}$$

H_1	Critical region
$\sigma^2 \neq \sigma_0^2$	$\chi^2_{\text{cal}} < \chi^2_{\alpha/2,\,n-1}$ or $\chi^2_{\text{cal}} > \chi^2_{1-\alpha/2,\,n-1}$
$\sigma^2 < \sigma_0^2$	$\chi^2_{\text{cal}} < \chi^2_{\alpha,\,n-1}$
$\sigma^2 > \sigma_0^2$	$\chi^2_{\text{cal}} > \chi^2_{1-\alpha,\,n-1}$

A claim about σ is tested as the same claim about σ^2: for example $\sigma > 2$ is $\sigma^2 > 4$.

Worked Example

Worked Example 2.4.2. A fund manager states that the standard deviation of the fund's monthly return is 2%. An investor suspects the fund is riskier. Over the last 24 months the standard deviation of the monthly return was $s = 2.6\%$ (hypothetical figures). Assume monthly returns are normally distributed.

(a) Construct a 95% confidence interval for the variance and the standard deviation of the monthly return.
(b) At $\alpha = 0.05$, test whether the standard deviation is more than 2%.

Solution

$s^2 = 2.6^2 = 6.76$, df $= 23$, $(n-1)s^2 = 23(6.76) = 155.48$.
(a) $\chi^2_{0.025,\,23} = 11.7$ and $\chi^2_{0.975,\,23} = 38.1$.

$$\left(\frac{155.48}{38.1},\ \frac{155.48}{11.7}\right) = (4.0808,\ 13.2889)$$

For σ: $(\sqrt{4.0808},\ \sqrt{13.2889}) = (2.0201,\ 3.6454)$
We are 95% confident that the standard deviation of the fund's monthly return is between 2.0201% and 3.6454%. Note that $s = 2.6$ is not in the middle of the interval.
(b)
Step 1: Hypotheses. $H_0 : \sigma^2 = 4$ $\quad$ $H_1 : \sigma^2 > 4$ (that is, $\sigma > 2$)

Step 2: Level of significance. $\alpha = 0.05$
Step 3: Test statistic.

$$\chi^2_{\text{cal}} = \frac{(n-1)s^2}{\sigma_0^2} = \frac{155.48}{4} = 38.870$$

Step 4: Critical value and critical region. df $= 23$: $\chi^2_{0.95,\,23} = 35.2$. Critical region: $\chi^2_{\text{cal}} > 35.2$.
Step 5: Decision and interpretation. $\chi^2_{\text{cal}} = 38.870$ falls in the critical region ($38.870 > 35.2$), so we reject H_0. That is, the standard deviation of the fund's monthly return is more than 2% at the 0.05 level of significance.

Your Turn

Your Turn 2.4.2

A price-stabilization policy aims to keep the standard deviation of the weekly retail price of pork at 3 baht per kg. Over 20 weeks the standard deviation was $s = 4.5$ baht (hypothetical figures). Assume prices are normally distributed.

(a) Construct a 95% confidence interval for σ.
(b) At $\alpha = 0.05$, test whether the standard deviation is different from 3 baht.

Step 1: Hypotheses

Step 2: Level of significance

Step 3: Test statistic

Step 4: Critical value and critical region

Step 5: Decision and interpretation

Formula Summary

	Formula
Condition	population normal
$(1-\alpha)100\%$ CI for σ^2	$\left(\sqrt{\dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}},\ \sqrt{\dfrac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}}}\right)$; square root for σ
Test statistic	$\chi^2_{\text{cal}} = \dfrac{(n-1)s^2}{\sigma_0^2}$, $\text{df} = n-1$
Critical region	$\neq$: $\chi^2_{\text{cal}} < \chi^2_{\alpha/2}$ or $> \chi^2_{1-\alpha/2}$; $<$: $\chi^2_{\text{cal}} < \chi^2_{\alpha}$; $>$: $\chi^2_{\text{cal}} > \chi^2_{1-\alpha}$

Exercises

2.4.1 Before a new ordering system, the standard deviation of a shop's daily sales was 10 units. After the system was introduced, 16 days of sales had a standard deviation of 7.2 units (hypothetical figures). At $\alpha = 0.05$, test whether the standard deviation is now less than 10 units. Assume normality.

2.4.2 The daily change in an exchange rate over 25 trading days has a standard deviation of 0.35 baht (hypothetical figures). Assuming normality, construct a 90% confidence interval for the standard deviation of the daily change.

2.4.3 Explain why the confidence interval for σ^2 is not of the form "estimate $\pm$ margin of error".

Solutions

Solution: Your Turn 2.4.1

(a) $\chi^2_{0.05,\,19} = 10.1$

(b) $\chi^2_{0.995,\,10} = 25.2$

(c) $\chi^2_{0.025,\,30} = 16.8$

Solution: Your Turn 2.4.2

$s^2 = 20.25$, df $= 19$, $(n-1)s^2 = 384.75$; $\chi^2_{0.025,\,19} = 8.91$, $\chi^2_{0.975,\,19} = 32.9$.

(a) CI for σ^2: $(384.75/32.9,\ 384.75/8.91) = (11.6945,\ 43.1818)$. CI for σ: (3.4197, 6.5713). We are 95% confident that the standard deviation of the weekly pork price is between 3.4197 and 6.5713 baht.

(b)

Step 1. $H_0 : \sigma^2 = 9 \quad H_1 : \sigma^2 \neq 9$

Step 2. $\alpha = 0.05$

Step 3. $\chi^2_{\text{cal}} = 384.75/9 = 42.750$

Step 4. Critical region: $\chi^2_{\text{cal}} < 8.91$ or $\chi^2_{\text{cal}} > 32.9$.

Step 5. $\chi^2_{\text{cal}} = 42.750$ falls in the critical region $(42.750 > 32.9)$, so we reject H_0. That is, the standard deviation of the weekly pork price is different from 3 baht at the 0.05 level of significance; since $s = 4.5$, it is higher than 3 baht, so the policy target was not met.

Solution: Exercise 2.4.1

Step 1. $H_0 : \sigma^2 = 100 \quad H_1 : \sigma^2 < 100$

Step 2. $\alpha = 0.05$

Step 3. $\chi^2_{\text{cal}} = 15(51.84)/100 = 7.776$

Step 4. $\chi^2_{0.05,\,15} = 7.26$. Critical region: $\chi^2_{\text{cal}} < 7.26$.

Step 5. $\chi^2_{\text{cal}} = 7.776$ does not fall in the critical region $(7.776 > 7.26)$, so we accept H_0. That is, the standard deviation of daily sales is equal to 10 units at the 0.05 level of significance.

Solution: Exercise 2.4.2

$(n-1)s^2 = 24(0.1225) = 2.94$; $\chi^2_{0.05,\,24} = 13.8$, $\chi^2_{0.95,\,24} = 36.4$. CI for σ^2: (0.0808, 0.2130); CI for σ: (0.2843, 0.4615). We are 90% confident that the standard deviation of the daily change is between 0.2843 and 0.4615 baht.

Solution: Exercise 2.4.3

The chi-square distribution is skewed to the right, so the two critical values are not the same distance from the centre. Dividing $(n-1)s^2$ by them gives limits that are not equally far from s^2.

3 Two-Sample Inference

3.1 Two Proportions

Learning Objectives

1. Construct and interpret a confidence interval for the difference $p_1 - p_2$.
2. Carry out a z test for $H_0 : p_1 = p_2$ using the pooled proportion.
3. Explain why the test pools the two samples but the CI does not.

Concept

We often compare two groups: urban and rural households, men and women in the labour force, customers of two brands. The parameter of interest is the difference $p_1 - p_2$, estimated by $\hat{p}_1 - \hat{p}_2$. For two independent random samples that are large enough, $\hat{p}_1 - \hat{p}_2$ is approximately normal with standard error

$$\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}.$$

The same idea as in Section 2.3 decides which values go into the standard error:

- **Confidence interval:** use $\hat{p}_1$ and $\hat{p}_2$ separately.
- **Hypothesis test:** H_0 says $p_1 = p_2$, so both groups share one proportion. Our best estimate of it combines the two samples, the **pooled proportion**

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}.$$

Formulas and Conditions

Conditions: two independent random samples; $n\hat{p} \geq 5$ and $n(1-\hat{p}) \geq 5$ in each sample.

$(1-\alpha)100\%$ **confidence interval for** $p_1 - p_2$

$$(\hat{p}_1 - \hat{p}_2) \pm z_{1-\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Test statistic for $H_0 : p_1 = p_2$

$$z_{\text{cal}} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

H_1	Critical region
$p_1 \neq p_2$	$z_{\text{cal}} < z_{\alpha/2}$ or $z_{\text{cal}} > z_{1-\alpha/2}$
$p_1 < p_2$	$z_{\text{cal}} < z_{\alpha}$
$p_1 > p_2$	$z_{\text{cal}} > z_{1-\alpha}$

Worked Example

Worked Example 3.1.1. A survey asks households whether they have any outstanding debt. In urban areas, 171 of 300 households have debt; in rural areas, 165 of 250 households have debt (hypothetical figures).

(a) Construct a 95% confidence interval for the difference between the urban and rural proportions.
(b) At $\alpha = 0.05$, test whether the proportion of households with debt is different in urban and rural areas.

Solution

Let group 1 = urban and group 2 = rural. $\hat{p}_1 = 171/300 = 0.57$, $\hat{p}_2 = 165/250 = 0.66$, $\hat{p}_1 - \hat{p}_2 = -0.09$.

(a)

$$\sqrt{\frac{0.57(0.43)}{300} + \frac{0.66(0.34)}{250}} = 0.0414$$

$$-0.09 \pm 1.960(0.0414) = -0.09 \pm 0.0811$$

95% CI: (-0.1711, -0.0089)

We are 95% confident that the difference between the urban and rural proportions of households with debt is between -0.1711 and -0.0089. Both limits are negative, so the urban proportion is lower.

(b)

Step 1: Hypotheses. $H_0 : p_1 = p_2 \quad H_1 : p_1 \neq p_2$

Step 2: Level of significance. $\alpha = 0.05$

Step 3: Test statistic.

$$\bar{p} = \frac{171 + 165}{300 + 250} = 0.6109$$

$$z_{\text{cal}} = \frac{-0.09}{\sqrt{0.6109(0.3891)\left(\frac{1}{300}+\frac{1}{250}\right)}} = \frac{-0.09}{0.0418} = -2.153$$

Step 4: Critical value and critical region. Critical region: $z_{\text{cal}} < -1.960$ or $z_{\text{cal}} > 1.960$.

Step 5: Decision and interpretation. $z_{\text{cal}} = -2.153$ falls in the critical region ($-2.153 < -1.960$), so we reject H_0. That is, the proportion of households with debt is different in urban and rural areas at the 0.05 level of significance; since $\hat{p}_1 < \hat{p}_2$, the proportion is lower in urban areas.

Your Turn

Your Turn 3.1.1

Two brands of instant noodles raised their prices by the same amount. Of 200 regular customers of Brand A, 58 switched to another brand; of 180 regular customers of Brand B, 36 switched (hypothetical figures).

(a) Construct a 95% confidence interval for $p_A - p_B$.
(b) At $\alpha = 0.05$, test whether the proportion of customers who switched is higher for Brand A.

Step 1: Hypotheses

Step 2: Level of significance

Step 3: Test statistic

Step 4: Critical value and critical region

Step 5: Decision and interpretation

Looking Back: Methods So Far

Parameter	Estimate	Distribution	Section
μ (σ known)	$\bar{x}$	z	2.1
μ (σ unknown)	$\bar{x}$	t, df $= n-1$	2.2
p	$\hat{p}$	z	2.3
σ^2	s^2	χ^2, df $= n-1$	2.4
$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$	z	3.1

Every test so far followed the same five steps and made the decision with the critical region. Section 3.2 introduces a second way to make the decision, the p-value, and shows how to read the results from R.

Formula Summary

	Confidence interval	Hypothesis test
Standard error	$\sqrt{\dfrac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \dfrac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	$\sqrt{\bar{p}(1-\bar{p})\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}$, $\bar{p} = \dfrac{x_1 + x_2}{n_1 + n_2}$
Formula	$(\hat{p}_1 - \hat{p}_2) \pm z_{1-\alpha/2}\,\mathrm{SE}$	$z_{\mathrm{cal}} = \dfrac{\hat{p}_1 - \hat{p}_2}{\mathrm{SE}}$

Exercises

3.1.1 In a labour force survey, 45 of 150 women and 62 of 160 men aged 15–24 were not in employment, education or training (hypothetical figures). Construct a 90% confidence interval for $p_{\text{women}} - p_{\text{men}}$.

3.1.2 A shop tests two price displays. With display 1, 84 of 240 visitors bought the product; with display 2, 96 of 240 bought it (hypothetical figures). At $\alpha = 0.05$, test whether display 1 has a lower purchase rate.

Solutions

Solution: Your Turn 3.1.1

$\hat{p}_A = 0.29$, $\hat{p}_B = 0.2$, $\hat{p}_A - \hat{p}_B = 0.09$.

(a) SE = 0.0438; $0.09 \pm 1.960(0.0438) = 0.09 \pm 0.0858$. 95% CI: (0.0042, 0.1758). We are 95% confident that the difference between the proportions of Brand A and Brand B customers who switched is between 0.0042 and 0.1758.

(b)

Step 1. $H_0 : p_A = p_B \quad H_1 : p_A > p_B$

Step 2. $\alpha = 0.05$

Step 3. $\bar{p} = (58 + 36)/(200 + 180) = 0.2474$; $z_{\text{cal}} = 0.09/0.0443 = 2.032$

Step 4. $z_{0.95} = 1.645$. Critical region: $z_{\text{cal}} > 1.645$.

Step 5. $z_{\text{cal}} = 2.032$ falls in the critical region ($2.032 > 1.645$), so we reject H_0. That is, the proportion of customers who switched is higher for Brand A than for Brand B at the 0.05 level of significance.

Solution: Exercise 3.1.1

$\hat{p}_1 = 0.3$, $\hat{p}_2 = 0.3875$, difference $= -0.0875$, SE $= 0.0537$, $z_{0.95} = 1.645$. -0.0875 ± 0.0883: 90% CI (-0.1758, 0.0008). We are 90% confident that the difference between the proportions of women and men not in employment, education or training is between -0.1758 and 0.0008. The interval contains 0.

Solution: Exercise 3.1.2

Step 1. $H_0 : p_1 = p_2 \quad H_1 : p_1 < p_2$

Step 2. $\alpha = 0.05$

Step 3. $\hat{p}_1 = 0.35$, $\hat{p}_2 = 0.4$, $\bar{p} = 0.375$; $z_{\text{cal}} = -0.05/0.0442 = -1.131$

Step 4. $z_{0.05} = -1.645$. Critical region: $z_{\text{cal}} < -1.645$.

Step 5. $z_{\text{cal}} = -1.131$ does not fall in the critical region ($-1.131 > -1.645$), so we accept H_0. That is, the purchase rates with the two displays are equal at the 0.05 level of significance.

3.2 P-values and R Output

Learning Objectives

1. Explain what a p-value is and use it to make the decision of a test.
2. Read the output of `t.test()` in R: test statistic, df, p-value, CI and estimate.
3. Check that R output agrees with a calculation by hand.
4. Choose the R output whose alternative hypothesis matches the research question.

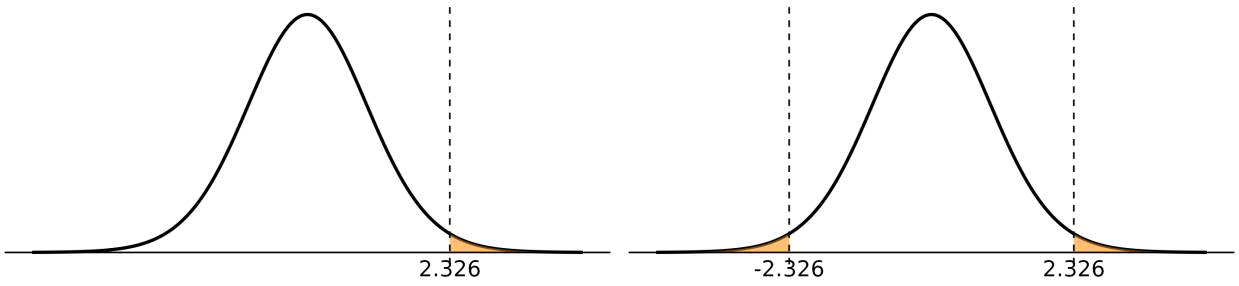
Concept

So far we have decided by comparing the test statistic with a **critical value**. Statistical software gives a second, equivalent way: the **p-value**.

p-value

The **p-value** is the probability, assuming H_0 is true, of getting a test statistic at least as extreme as the one we observed, in the direction(s) of H_1.
A small p-value means our sample would be unusual if H_0 were true.
Decision rule: reject H_0 if p-value $< \alpha$; accept H_0 if p-value $\geq \alpha$.

Example. In Worked Example 2.2.2 (rider earnings, $H_1 : \mu > 600$), $t_{\text{cal}} = 2.326$ with df $= 24$. The p-value is the area to the **right** of 2.326 under the t curve, because H_1 points to the right. It is about 0.0144.



(a) Right-sided H1: area to the right of t_cal (b) Two-sided H1: both tails, twice the one-tail area

Figure 3.1: The p-value as an area under the t curve with df = 24

H_1	p-value
$>$	area to the **right** of the test statistic
$<$	area to the **left** of the test statistic
$\neq$	**twice** the area beyond the test statistic, in both tails

The critical-value method and the p-value method always give the **same decision**. The critical region is the set of test statistics whose p-value is less than α. The printed tables do not give exact p-values for t, χ^2 or F, so we read p-values from software.

> What the p-value is not
>
> The p-value is **not** the probability that H_0 is true. It is calculated assuming H_0 is true.

Reading R Output

R is free statistical software. In this course you do not need to write R code, but you must be able to **read** its output. The function `t.test()` carries out the t test for one mean. For the riders in Worked Example 2.2.2 we use `t.test(earnings, mu = 600, alternative = "greater")`:

```
    One Sample t-test

data:  earnings
t = 2.3259, df = 24, p-value = 0.0144
alternative hypothesis: true mean is greater than 600
95 percent confidence interval:
 611.8887      Inf
sample estimates:
mean of x
   644.96
```

Line in the output	Meaning	Hand calculation (2.2.2)
`t =`	test statistic t_{cal}	2.326
`df =`	degrees of freedom $n-1$	24
`p-value =`	p-value	(from the table: t_{cal} in the critical region)
`alternative hypothesis:`	H_1 used by R	$H_1 : \mu > 600$
`mean of x`	$\bar{x}$	644.96

The numbers match the hand calculation. Now the decision can be written with the p-value:

> p-value $= 0.0144 < \alpha = 0.05$, so we reject H_0. That is, the mean daily earnings of riders in Chiang Mai are more than 600 baht at the 0.05 level of significance.

The confidence interval in one-sided output

With `alternative = "greater"` or `"less"`, R prints a **one-sided** interval with `Inf` or `-Inf`. In this course we use the **two-sided** CI, which R prints with the default two-sided test:

```
    One Sample t-test

data:  earnings
t = 2.3259, df = 24, p-value = 0.0288
alternative hypothesis: true mean is not equal to 600
95 percent confidence interval:
 605.0649 684.8551
sample estimates:
mean of x
   644.96
```

The 95% CI (605.06, 684.86) is the same as in Worked Example 2.2.2. The test statistic does not change, but the two-sided p-value (0.0288) is twice the one-sided p-value.

Reading numbers in R

- R may print `p-value = 2.1e-05`. This means $2.1 \times 10^{-5} = 0.000021$.
- `p-value < 2.2e-16` means the p-value is smaller than any number R can print; report it as p-value < 0.0001.
- In this book we round the test statistic to 3 decimals and the p-value to 4 decimals.

The Five Steps with a p-value

From this section on, Step 4 is the p-value instead of the critical value:

1. **Hypotheses**
2. **Level of significance**
3. **Test statistic** (from the output, or calculated by hand)
4. **p-value** (from the output)
5. **Decision and interpretation**: p-value = … [$<$, $\geq$] $\alpha = 0.05$, so we [reject / accept] H_0. That is, …

Worked Example

Worked Example 3.2.1. An energy agency records the monthly electricity bill (baht) of 30 randomly selected households (hypothetical data). It wants to know whether the mean bill is different from 1,200 baht. R gives:

```
    One Sample t-test

data:  bill
t = 1.6945, df = 29, p-value = 0.1009
alternative hypothesis: true mean is not equal to 1200
95 percent confidence interval:
 1187.180 1336.686
sample estimates:
mean of x
 1261.933
```

Use the output to (a) write the 95% CI and (b) test the hypothesis at $\alpha = 0.05$.

Solution

(a) We are 95% confident that the mean monthly electricity bill is between 1,187.18 and 1,336.69 baht.
(b)
Step 1: Hypotheses. $H_0 : \mu = 1200 \quad H_1 : \mu \neq 1200$
Step 2: Level of significance. $\alpha = 0.05$
Step 3: Test statistic. $t_{\text{cal}} = 1.694$, df $= 29$
Step 4: p-value. p-value $= 0.1009$
Step 5: Decision and interpretation. p-value $= 0.1009 \geq \alpha = 0.05$, so we accept H_0. That is, the mean monthly electricity bill is equal to 1,200 baht at the 0.05 level of significance.
The CI contains 1,200, which agrees with accepting H_0 in a two-sided test.

Your Turn

Your Turn 3.2.1

For the rider earnings data, a labour researcher tests whether the mean daily earnings are different from 620 baht. Use the output below to test the hypothesis at $\alpha = 0.05$ in five steps.

```
    One Sample t-test

data:  earnings
t = 1.2913, df = 24, p-value = 0.2089
alternative hypothesis: true mean is not equal to 620
95 percent confidence interval:
 605.0649 684.8551
sample estimates:
mean of x
   644.96
```

Formula Summary

p-value	probability, assuming H_0 is true, of a test statistic at least as extreme as observed, in the direction of H_1
Decision	reject H_0 if p-value $< \alpha$; accept H_0 if p-value $\geq \alpha$
One mean in R	`t.test(x, mu = mu0, alternative = "two.sided" / "less" / "greater")`
CI in R	read from the two-sided output

Exercises

3.2.1 The electricity agency in Worked Example 3.2.1 now asks whether the mean bill is **more than** 1,200 baht. Four outputs are shown below.

```
Output A


    One Sample t-test

data:  bill
t = 1.6945, df = 29, p-value = 0.1009
alternative hypothesis: true mean is not equal to 1200
95 percent confidence interval:
 1187.180 1336.686
sample estimates:
mean of x
 1261.933


Output B
```

```
    One Sample t-test

data:  bill
t = 1.6945, df = 29, p-value = 0.05044
alternative hypothesis: true mean is greater than 1200
95 percent confidence interval:
 1199.83     Inf
sample estimates:
mean of x
 1261.933
```

Output C

```
    One Sample t-test

data:  bill
t = 1.6945, df = 29, p-value = 0.9496
alternative hypothesis: true mean is less than 1200
95 percent confidence interval:
     -Inf 1324.036
sample estimates:
mean of x
 1261.933
```

Output D

```
    One Sample t-test

data:  bill
t = 0.32649, df = 29, p-value = 0.3732
alternative hypothesis: true mean is greater than 1250
95 percent confidence interval:
 1199.83     Inf
sample estimates:
mean of x
 1261.933
```

(a) Which output matches the agency's question? Explain.
(b) Write the decision and interpretation at $\alpha = 0.05$.
(c) Why is the test statistic the same in Outputs A, B and C, but not in D?

3.2.2 For each p-value, write the decision at $\alpha = 0.05$: (a) 0.0312, (b) 0.0500, (c) 0.1847, (d) `5.4e-04`.

3.2.3 A student says: "The p-value is 0.03, so there is a 3% chance that H_0 is true." Explain what is wrong.

Solutions

Solution: Your Turn 3.2.1

Step 1. $H_0 : \mu = 620 \quad H_1 : \mu \neq 620$

Step 2. $\alpha = 0.05$

Step 3. $t_{\text{cal}} = 1.291$, df $= 24$

Step 4. p-value $= 0.2089$

Step 5. p-value $= 0.2089 \geq \alpha = 0.05$, so we accept H_0. That is, the mean daily earnings of riders are equal to 620 baht at the 0.05 level of significance. (This agrees with Section 2.2: 620 is inside the 95% CI.)

Solution: Exercise 3.2.1

(a) Output B: $\mu_0 = 1200$ and the alternative hypothesis is "greater than 1200", which matches $H_1 : \mu > 1200$.

(b) p-value $= 0.0504 \geq \alpha = 0.05$, so we accept H_0. That is, the mean monthly electricity bill is equal to 1,200 baht at the 0.05 level of significance.

(c) The test statistic depends on μ_0 but not on H_1. A, B and C all use $\mu_0 = 1200$; D uses $\mu_0 = 1250$. Only the p-value depends on H_1.

Solution: Exercise 3.2.2

(a) $0.0312 < 0.05$: reject H_0.
(b) $0.0500 \geq 0.05$: accept H_0.
(c) $0.1847 \geq 0.05$: accept H_0.
(d) `5.4e-04` $= 0.00054 < 0.05$: reject H_0.

Solution: Exercise 3.2.3

The p-value is calculated assuming H_0 is true. It is the probability of a result at least as extreme as the sample result if H_0 were true, not the probability that H_0 is true.

3.3 Paired Means

Learning Objectives

1. Recognize paired data from the way the data were collected.
2. Reduce paired data to differences d and use the one-sample t method on them.
3. Construct a CI and carry out a test for the mean difference μ_d, by hand and from R.

Concept

Data are **paired** when each observation in one group is matched with one observation in the other group. Common designs:

- **Before and after:** the same households are measured before and after a policy.
- **Matched units:** the same products are priced at two shops; twin regions are compared.

Each pair gives one **difference** $d = x_1 - x_2$ (for example, after − before). The pairing removes the variation between units (rich and poor households differ a lot, but each is compared with itself), so we analyse the differences only. A paired test is a **one-sample t test on** d (Section 2.2) with $H_0 : \mu_d = 0$.

> Paired or independent?
>
> Decide from **how the data were collected**, not from the numbers:
>
> - the same units measured twice, or units matched one-to-one → **paired**
> - two separate random samples, with no link between a unit in one group and a unit in the other → **independent** (Section 3.5)
>
> Paired samples always have the same size. Equal sample sizes alone do not make the data paired.

Formulas and Conditions

Conditions: pairs are a random sample; the differences are approximately normal.

With n pairs, $\bar{d}$ the mean and s_d the standard deviation of the differences:

$(1-\alpha)100\%$ **confidence interval for** μ_d

$$\bar{d} \pm t_{1-\alpha/2,\, n-1}\, \frac{s_d}{\sqrt{n}}$$

Test statistic for $H_0 : \mu_d = 0$, with $\text{df} = n - 1$

$$t_{\text{cal}} = \frac{\bar{d}}{s_d/\sqrt{n}}$$

In R: `t.test(after, before, paired = TRUE, alternative = ...)` uses $d = \text{after} - \text{before}$ (first argument minus second).

Worked Example

Worked Example 3.3.1. The government gave a cash transfer to low-income households. A researcher recorded the monthly spending (baht) of 10 households in the month before and the month after the transfer (hypothetical data). Did mean spending increase?

Table 3.1: Monthly spending (baht) before and after the transfer, and the difference d = after - before

Household	Before	After	d
1	14,570	16,140	1,570
2	12,550	14,740	2,190
3	14,270	14,840	570
4	12,370	14,220	1,850
5	12,130	12,960	830
6	12,420	13,790	1,370
7	15,090	16,590	1,500
8	16,280	17,840	1,560
9	15,050	14,380	-670
10	13,610	13,280	-330

(a) Construct a 95% confidence interval for the mean change in spending.
(b) At $\alpha = 0.05$, test whether mean spending increased after the transfer.

Solution

The same households are measured twice, so the data are paired. From the differences: $n = 10$, $\bar{d} = 1044$, $s_d = 937.8604$.

$$\frac{s_d}{\sqrt{n}} = \frac{937.8604}{\sqrt{10}} = 296.5775$$

(a) $t_{0.975,\,9} = 2.262$:

$$1044 \pm 2.262(296.5775) = 1044 \pm 670.8583$$

95% CI: (373.14, 1,714.86)

We are 95% confident that the mean increase in monthly spending is between 373.14 and 1,714.86 baht.

(b)

Step 1: Hypotheses. $H_0 : \mu_d = 0 \quad H_1 : \mu_d > 0$ (where $d = \text{after} - \text{before}$)

Step 2: Level of significance. $\alpha = 0.05$

Step 3: Test statistic.

$$t_{\text{cal}} = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{1044}{296.5775} = 3.520$$

Step 4: p-value. From the R output below, p-value = 0.0033.

Step 5: Decision and interpretation. p-value = $0.0033 < \alpha = 0.05$, so we reject H_0. That is, mean monthly spending increased after the cash transfer at the 0.05 level of significance.

Reading R Output

```
    Paired t-test

data:  after and before
t = 3.5202, df = 9, p-value = 0.003257
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 500.3399      Inf
sample estimates:
mean difference
           1044
```

`t =` 3.520 and `df =` 9 match the hand calculation, and `mean difference` is $\bar{d} = 1044$. For the 95% CI we use the two-sided output:

```
    Paired t-test

data:  after and before
t = 3.5202, df = 9, p-value = 0.006514
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
  373.0951 1714.9049
sample estimates:
mean difference
           1044
```

R gives (373.10, 1,714.90). The hand calculation gave (373.14, 1,714.86). The small difference comes from rounding: the table gives $t_{0.975,\,9} = 2.262$, while R uses 2.262157.

Your Turn

Your Turn 3.3.1

The daily wage (baht) of 12 workers was recorded before and after a training course (hypothetical data). For $d = \text{after} - \text{before}$, $\bar{d} = 7.6667$ and $s_d = 24.4776$.

Table 3.2

Worker	Before	After	d
1	371	388	17
2	405	460	55
3	409	425	16
4	538	562	24
5	357	315	-42
6	351	379	28
7	414	413	-1
8	416	412	-4
9	390	398	8
10	438	418	-20
11	401	401	0
12	428	439	11

(a) Calculate t_{cal} by hand.
(b) Use the R output to test whether the training changed the mean daily wage at $\alpha = 0.05$.

```
    Paired t-test

data:  wage_after and wage_before
t = 1.085, df = 11, p-value = 0.3011
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -7.88564 23.21897
sample estimates:
mean difference
       7.666667
```

Formula Summary

	Formula
Differences	d = first − second for each pair; $\bar{d}$, s_d, n pairs
$(1-\alpha)100\%$ CI for μ_d	$\bar{d} \pm t_{1-\alpha/2,\, n-1}\, s_d/\sqrt{n}$
Test statistic	$t_{\text{cal}} = \dfrac{\bar{d}}{s_d/\sqrt{n}}$, $\text{df} = n-1$
R	`t.test(x1, x2, paired = TRUE, alternative = ...)`, d = `x1 − x2`

Exercises

3.3.1 Paired or independent?

(a) The household income of 50 families is recorded in 2024 and again in 2025.
(b) The wages of 40 randomly chosen workers in Chiang Mai are compared with the wages of 40 randomly chosen workers in Lampang.
(c) The same 20 products are priced at a supermarket and at a fresh market.
(d) 30 firms that adopted a new technology are compared with 30 other firms that did not.

3.3.2 A consumer group priced the same 8 products at Shop A and Shop B (hypothetical data). It wants to know whether prices at Shop A are lower on average. Use the output to write the decision and interpretation at $\alpha = 0.05$.

```
    Paired t-test

data:  shopA and shopB
t = -4.1501, df = 7, p-value = 0.002147
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
     -Inf -1.56252
sample estimates:
mean difference
         -2.875
```

Solutions

Solution: Your Turn 3.3.1

(a) $s_d/\sqrt{n} = 24.4776/\sqrt{12} = 7.0661$; $t_{\text{cal}} = 7.6667/7.0661 = 1.085$ (R: 1.085).

(b)

Step 1. $H_0 : \mu_d = 0 \quad H_1 : \mu_d \neq 0$

Step 2. $\alpha = 0.05$

Step 3. $t_{\text{cal}} = 1.085$, df $= 11$

Step 4. p-value $= 0.3011$

Step 5. p-value $= 0.3011 \geq \alpha = 0.05$, so we accept H_0. That is, the training did not change the mean daily wage (the mean difference is equal to 0) at the 0.05 level of significance.

Solution: Exercise 3.3.1

(a) Paired (same families twice). (b) Independent (two separate samples).
(b) Paired (same products at two places). (d) Independent (different firms).

Solution: Exercise 3.3.2

$H_0 : \mu_d = 0 \quad H_1 : \mu_d < 0$ (d = Shop A − Shop B). $t_{\text{cal}} = -4.150$, df $= 7$, p-value $= 0.0021$. p-value $= 0.0021 < \alpha = 0.05$, so we reject H_0. That is, prices at Shop A are lower than at Shop B on average at the 0.05 level of significance.

3.4 Two Variances

Learning Objectives

1. Read the F table (Table 7), including lower-tail values from the reciprocal rule.
2. Construct and interpret a confidence interval for the ratio σ_1^2/σ_2^2.
3. Test whether two population variances are equal, by hand and with `var.test()` in R.

Concept

Is one investment riskier than another? Are prices more stable in one market than in another? These questions compare two variances. We compare them with a **ratio**, not a difference:

$$F = \frac{s_1^2}{s_2^2}.$$

If the two populations are normal with equal variances, this ratio follows an F **distribution** with two degrees of freedom: $\text{df}_1 = n_1 - 1$ (numerator) and $\text{df}_2 = n_2 - 1$ (denominator). Like χ^2, the F distribution is positive and skewed to the right.

The F test is also the starting point for the next two topics: Section 3.5 uses it to choose between two t tests, and Chapter 4 (ANOVA) is built on an F ratio.

Reading the F Table

Table 7 gives F values with area **0.90, 0.95, 0.975, 0.99 or 0.995** to the left, one page for each area. Columns are df_1 (numerator) and rows are df_2 (denominator). The order of the df matters.

Areas such as 0.025 and 0.05 are not printed. Use the **reciprocal rule**, which swaps the two df:

$$F_{\alpha,\,\mathrm{df}_1,\,\mathrm{df}_2} = \frac{1}{F_{1-\alpha,\,\mathrm{df}_2,\,\mathrm{df}_1}}$$

Worked Example 3.4.1. Find (a) $F_{0.95,\,4,\,10}$, (b) $F_{0.975,\,12,\,15}$, (c) $F_{0.025,\,12,\,15}$.

Solution

(a) Page for 0.95, column 4, row 10: $F_{0.95,\,4,\,10} = 3.48$.

(b) Page for 0.975, column 12, row 15: $F_{0.975,\,12,\,15} = 2.96$.

(c) Swap the df and use the 0.975 page: $F_{0.025,\,12,\,15} = \dfrac{1}{F_{0.975,\,15,\,12}} = \dfrac{1}{3.18} = 0.3145$.

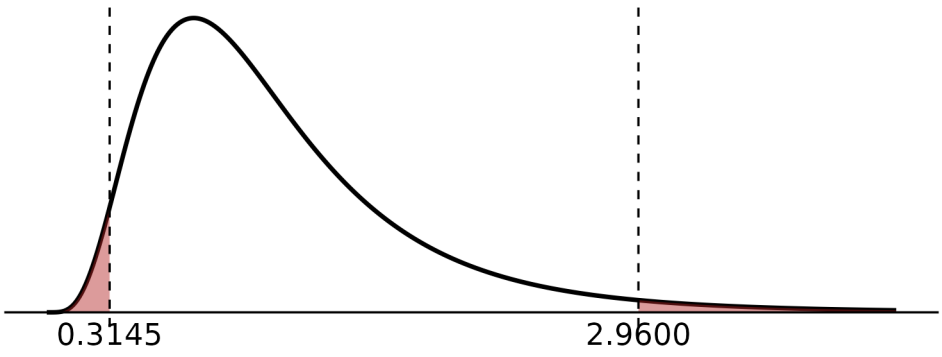
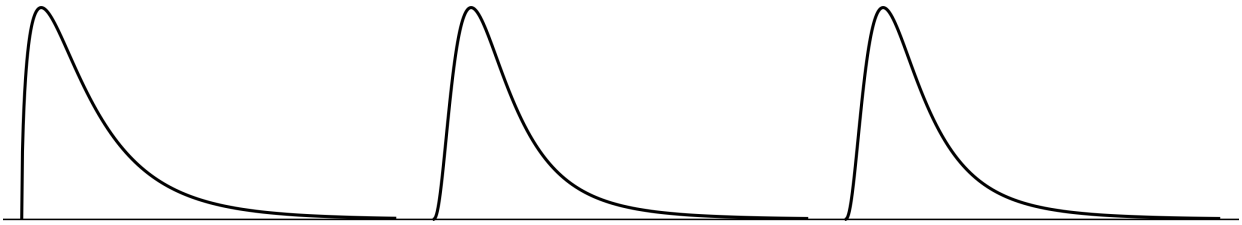


Figure 3.2: Areas 0.025 in each tail of the F distribution with df = (12, 15)

Your Turn 3.4.1

Shade the area to the left on each curve, then find (a) $F_{0.95,\,3,\,20}$, (b) $F_{0.975,\,8,\,10}$, (c) $F_{0.025,\,8,\,10}$.



(a) (b) (c)

Figure 3.3: Shade the area to the left before reading Table 7

Formulas and Conditions

Conditions: two independent random samples from **normal** populations.

$(1-\alpha)100\%$ **confidence interval for** σ_1^2/σ_2^2

$$\left(\frac{s_1^2/s_2^2}{F_{1-\alpha/2,\,\mathrm{df}_1,\,\mathrm{df}_2}},\ \frac{s_1^2/s_2^2}{F_{\alpha/2,\,\mathrm{df}_1,\,\mathrm{df}_2}}\right) = \left(\frac{s_1^2/s_2^2}{F_{1-\alpha/2,\,\mathrm{df}_1,\,\mathrm{df}_2}},\ \frac{s_1^2}{s_2^2}\,F_{1-\alpha/2,\,\mathrm{df}_2,\,\mathrm{df}_1}\right)$$

If the interval contains 1, the data are consistent with equal variances.

Test statistic for $H_0 : \sigma_1^2 = \sigma_2^2$, with $\mathrm{df} = (n_1 - 1,\ n_2 - 1)$

$$F_{\mathrm{cal}} = \frac{s_1^2}{s_2^2}$$

H_1	Critical region
$\sigma_1^2 \neq \sigma_2^2$	$F_{\mathrm{cal}} < F_{\alpha/2,\,\mathrm{df}_1,\,\mathrm{df}_2}$ or $F_{\mathrm{cal}} > F_{1-\alpha/2,\,\mathrm{df}_1,\,\mathrm{df}_2}$
$\sigma_1^2 < \sigma_2^2$	$F_{\mathrm{cal}} < F_{\alpha,\,\mathrm{df}_1,\,\mathrm{df}_2}$
$\sigma_1^2 > \sigma_2^2$	$F_{\mathrm{cal}} > F_{1-\alpha,\,\mathrm{df}_1,\,\mathrm{df}_2}$

In R: `var.test(x1, x2, alternative = ...)` uses $F = s_1^2/s_2^2$ with `x1` in the numerator.

Worked Example

Worked Example 3.4.2. An investor compares the risk of two mutual funds using their monthly returns (%): 13 months for Fund A and 16 months for Fund B (hypothetical data). The sample variances are $s_A^2 = 6.3586$ and $s_B^2 = 1.2101$. Assume returns are normally distributed.

(a) Construct a 95% confidence interval for σ_A^2/σ_B^2.
(b) At $\alpha = 0.05$, test whether the two funds have different variances (risk).

Solution

$\text{df}_1 = 13 - 1 = 12$, $\text{df}_2 = 16 - 1 = 15$, $s_A^2/s_B^2 = 6.3586/1.2101 = 5.255$.
(a) $F_{0.975,\,12,\,15} = 2.96$ and $F_{0.975,\,15,\,12} = 3.18$:

$$\left(\frac{5.255}{2.96},\ 5.255 \times 3.18\right) = (1.7753,\ 16.7109)$$

We are 95% confident that the ratio σ_A^2/σ_B^2 is between 1.7753 and 16.7109. The interval does not contain 1.
(b)
Step 1: Hypotheses. $H_0 : \sigma_A^2 = \sigma_B^2 \quad H_1 : \sigma_A^2 \neq \sigma_B^2$
Step 2: Level of significance. $\alpha = 0.05$
Step 3: Test statistic.

$$F_{\text{cal}} = \frac{s_A^2}{s_B^2} = \frac{6.3586}{1.2101} = 5.255, \qquad \text{df} = (12,\ 15)$$

Step 4: p-value. From the R output below, p-value = 0.0034. (With the table: critical region $F_{\text{cal}} < 0.3145$ or $F_{\text{cal}} > 2.96$.)
Step 5: Decision and interpretation. p-value = $0.0034 < \alpha = 0.05$, so we reject H_0. That is, the variances of the monthly returns of the two funds are different at the 0.05 level of significance; since $s_A^2 > s_B^2$, Fund A is riskier.

Reading R Output

```
    F test to compare two variances

data:  fund_A and fund_B
F = 5.2548, num df = 12, denom df = 15, p-value = 0.003424
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
  1.773311 16.695624
sample estimates:
```

```
ratio of variances
          5.254821
```

Output	Meaning
`F =`	$F_{\text{cal}} = s_1^2/s_2^2$
`num df, denom df`	$\text{df}_1 = n_1 - 1$, $\text{df}_2 = n_2 - 1$
`95 percent confidence interval`	CI for σ_1^2/σ_2^2
`ratio of variances`	s_1^2/s_2^2

R's CI (1.7733, 16.6956) differs a little from the hand calculation because R uses unrounded F values.

Your Turn

Your Turn 3.4.2

An analyst compares the daily price changes (%) of two stocks. For Stock X, $n = 11$ days and $s = 2.8$; for Stock Y, $n = 21$ days and $s = 1.9$ (hypothetical figures). Assume normality. At $\alpha = 0.05$, test whether Stock X is more volatile (has a larger variance) than Stock Y, using the critical region from Table 7.

Formula Summary

	Formula
Conditions	independent samples; both populations normal
Reciprocal rule	$F_{\alpha,\,\mathrm{df}_1,\,\mathrm{df}_2} = 1/F_{1-\alpha,\,\mathrm{df}_2,\,\mathrm{df}_1}$
$(1-\alpha)100\%$ CI for σ_1^2/σ_2^2	$\left(\dfrac{s_1^2/s_2^2}{F_{1-\alpha/2,\,\mathrm{df}_1,\,\mathrm{df}_2}},\ \dfrac{s_1^2}{s_2^2}F_{1-\alpha/2,\,\mathrm{df}_2,\,\mathrm{df}_1}\right)$
Test statistic	$F_{\mathrm{cal}} = s_1^2/s_2^2$, $\mathrm{df} = (n_1 - 1,\ n_2 - 1)$
R	`var.test(x1, x2, alternative = ...)`

Exercises

3.4.1 Find (a) $F_{0.99,\,5,\,12}$, (b) $F_{0.05,\,6,\,20}$, (c) $F_{0.975,\,24,\,30}$.

3.4.2 A regulator compares the interest rates (%) on personal loans from two groups of lenders: 16 loans from group A and 21 from group B (hypothetical data). It wants to know whether rates vary more in group A. Use the output to write the decision and interpretation at $\alpha = 0.05$.

```
    F test to compare two variances

data:  rate_bankA and rate_bankB
F = 3.5292, num df = 15, denom df = 20, p-value = 0.004783
alternative hypothesis: true ratio of variances is greater than 1
95 percent confidence interval:
 1.6018    Inf
sample estimates:
ratio of variances
          3.529204
```

Solutions

Solution: Your Turn 3.4.1

(a) $F_{0.95,\,3,\,20} = 3.1$

(b) $F_{0.975,\,8,\,10} = 3.85$

(c) $F_{0.025,\,8,\,10} = 1/F_{0.975,\,10,\,8} = 1/4.3 = 0.2326$

Solution: Your Turn 3.4.2

Step 1. $H_0 : \sigma_X^2 = \sigma_Y^2 \quad H_1 : \sigma_X^2 > \sigma_Y^2$

Step 2. $\alpha = 0.05$

Step 3. $F_{\text{cal}} = 2.8^2/1.9^2 = 7.84/3.61 = 2.172$, df = (10, 20)

Step 4. $F_{0.95,\,10,\,20} = 2.35$. Critical region: $F_{\text{cal}} > 2.35$.

Step 5. $F_{\text{cal}} = 2.172$ does not fall in the critical region ($2.172 < 2.35$), so we accept H_0. That is, the variances of the daily price changes of the two stocks are equal at the 0.05 level of significance.

Solution: Exercise 3.4.1

(a) 5.06 (b) $1/F_{0.95,\,20,\,6} = 1/3.87 = 0.2584$ (c) 2.14

Solution: Exercise 3.4.2

$H_0 : \sigma_A^2 = \sigma_B^2 \quad H_1 : \sigma_A^2 > \sigma_B^2$. $F_{\text{cal}} = 3.529$, df = (15, 20), p-value = $0.0048 < \alpha = 0.05$, so we reject H_0. That is, interest rates vary more in group A than in group B at the 0.05 level of significance.

3.5 Two Independent Means

Learning Objectives

1. Tell independent samples from paired samples by how the data were collected.
2. Use the F test to choose between the pooled t test and the Welch t test.
3. Carry out the pooled t test and CI by hand, and read both tests from R.

Concept

Two **independent** random samples are drawn from two populations, for example stores using two pricing strategies, or graduates of two universities. We estimate $\mu_1 - \mu_2$ by $\bar{x}_1 - \bar{x}_2$. Its standard error depends on whether the two population variances are equal:

- **Equal variances** ($\sigma_1^2 = \sigma_2^2$): combine the two sample variances into one **pooled variance** s_p^2 and use the **pooled t test** with df $= n_1 + n_2 - 2$. This is calculated by hand.
- **Unequal variances**: use the **Welch t test**. Its df is not a whole number, so we read it from R.

So the method has a first step: compare the variances with the F test of Section 3.4.

Choosing the method

1. Paired or independent? (Section 3.3) If paired, use the paired t test.
2. If independent, test $H_0 : \sigma_1^2 = \sigma_2^2$ (two-sided, $\alpha = 0.05$).
   - accept $H_0 \rightarrow$ **pooled t test**: `t.test(x1, x2, var.equal = TRUE)`
   - reject $H_0 \rightarrow$ **Welch t test**: `t.test(x1, x2)`

Formulas and Conditions

Conditions: two independent random samples; both populations approximately normal.

Pooled method (equal variances), df $= n_1 + n_2 - 2$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \qquad \text{SE} = \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$\text{CI: } (\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2,\, n_1+n_2-2}\, \text{SE} \qquad t_{\text{cal}} = \frac{\bar{x}_1 - \bar{x}_2}{\text{SE}}$$

Welch method (unequal variances): read t_{cal}, df, p-value and CI from `t.test(x1, x2)`.

H_1	Critical region (pooled)
$\mu_1 \neq \mu_2$	$t_{\text{cal}} < t_{\alpha/2,\,\text{df}}$ or $t_{\text{cal}} > t_{1-\alpha/2,\,\text{df}}$
$\mu_1 < \mu_2$	$t_{\text{cal}} < t_{\alpha,\,\text{df}}$
$\mu_1 > \mu_2$	$t_{\text{cal}} > t_{1-\alpha,\,\text{df}}$

Worked Example: Pooled t Test

Worked Example 3.5.1. A retail chain tries two pricing strategies. 12 stores use a **discount** strategy and 10 other stores use a **bundle** strategy. Daily sales (units) are recorded (hypothetical data):

Table 3.3: Daily sales (units) under two pricing strategies

Strategy	Sales	n	Mean	Variance
Discount	128, 148, 102, 121, 116, 116, 108, 142, 138, 128, 102, 136	12	123.75	239.2955
Bundle	119, 124, 109, 133, 91, 113, 106, 105, 112, 104	10	111.6	136.9333

(a) At $\alpha = 0.05$, check whether the variances are equal.
(b) At $\alpha = 0.05$, test whether mean sales are higher with the discount strategy.
(c) Construct a 95% confidence interval for $\mu_D - \mu_B$.

Solution

The stores are different stores, so the samples are independent.
(a) $H_0 : \sigma_D^2 = \sigma_B^2$ vs $H_1 : \sigma_D^2 \neq \sigma_B^2$. $F_{\text{cal}} = 239.2955/136.9333 = 1.748$; from `var.test()` below, p-value $= 0.4110 \geq 0.05$, so we accept H_0: the variances are equal. Use the **pooled** method.
(b)
Step 1: Hypotheses. $H_0 : \mu_D = \mu_B \quad H_1 : \mu_D > \mu_B$
Step 2: Level of significance. $\alpha = 0.05$
Step 3: Test statistic.

$$s_p^2 = \frac{11(239.2955) + 9(136.9333)}{12 + 10 - 2} = 193.2325$$

$$\text{SE} = \sqrt{193.2325\left(\frac{1}{12} + \frac{1}{10}\right)} = 5.952$$

$$t_{\text{cal}} = \frac{123.75 - 111.6}{5.952} = 2.041, \qquad \text{df} = 20$$

Step 4: p-value. From the R output, p-value $= 0.0273$. (With the table: $t_{0.95,\,20} = 1.725$, critical region $t_{\text{cal}} > 1.725$.)

Step 5: Decision and interpretation. p-value $= 0.0273 < \alpha = 0.05$, so we reject H_0. That is, mean daily sales are higher with the discount strategy than with the bundle strategy at the 0.05 level of significance.
(c) $t_{0.975,\,20} = 2.086$: $12.15 \pm 2.086(5.952) = 12.15 \pm 12.4159$. 95% CI: (-0.27, 24.57). We are 95% confident that the difference in mean daily sales (discount − bundle) is between -0.27 and 24.57 units.

One-sided test, two-sided interval

The one-sided test in (b) rejects H_0, but the 95% CI in (c) contains 0. This is not a contradiction: the CI matches a **two-sided** test, whose p-value (0.0546) is twice the one-sided p-value.

Reading R Output

```
    F test to compare two variances

data:  discount and bundle
F = 1.7475, num df = 11, denom df = 9, p-value = 0.411
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.4467023 6.2699696
sample estimates:
ratio of variances
          1.747533


    Two Sample t-test

data:  discount and bundle
t = 2.0413, df = 20, p-value = 0.02732
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 1.884527      Inf
sample estimates:
mean of x mean of y
   123.75    111.60


    Two Sample t-test
```

```
data:  discount and bundle
t = 2.0413, df = 20, p-value = 0.05463
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.2655939 24.5655939
sample estimates:
mean of x mean of y
   123.75    111.60
```

With `var.equal = TRUE` R prints **Two Sample t-test** (pooled); without it R prints **Welch Two Sample t-test**. Check the title before reading the numbers.

Worked Example: Welch t Test

Worked Example 3.5.2. A consumer researcher records the price (baht) of the same model of headphones at 15 online stores and 12 physical stores (hypothetical data). Is the mean price different?

```
    F test to compare two variances

data:  online and physical
F = 7.2367, num df = 14, denom df = 11, p-value = 0.002223
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
  2.154537 22.394556
sample estimates:
ratio of variances
           7.23668
```

```
    Welch Two Sample t-test

data:  online and physical
t = -5.5038, df = 18.55, p-value = 2.843e-05
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -309.7842 -138.8825
sample estimates:
mean of x mean of y
 1860.667  2085.000
```

Solution

Equal variances? p-value of the F test $= 0.0022 < 0.05$, so we reject $H_0 : \sigma_1^2 = \sigma_2^2$. Online prices vary much more. Use the **Welch** test (the second output).
Step 1: Hypotheses. $H_0 : \mu_{\text{online}} = \mu_{\text{physical}}$ $\quad$ $H_1 : \mu_{\text{online}} \neq \mu_{\text{physical}}$
Step 2: Level of significance. $\alpha = 0.05$
Step 3: Test statistic. $t_{\text{cal}} = -5.504$, df $= 18.55$
Step 4: p-value. p-value $=$ 2.843e-05 $(= 0.000028)$, which is < 0.0001
Step 5: Decision and interpretation. p-value $< \alpha = 0.05$, so we reject H_0. That is, the mean price is different online and in physical stores at the 0.05 level of significance; since $\bar{x}_{\text{online}} = 1{,}860.67 < \bar{x}_{\text{physical}} = 2{,}085.00$, it is lower online.
We are 95% confident that the difference in mean price (online − physical) is between -309.78 and -138.88 baht.

Your Turn

Your Turn 3.5.1

The starting monthly salaries (thousand baht) of 10 economics graduates from University A and 12 from University B are compared (hypothetical data): $\bar{x}_A = 24.01$, $s_A^2 = 12.8677$, $\bar{x}_B = 21.6083$, $s_B^2 = 6.8936$.

(a) Use the first output to decide which t test to use.
(b) Calculate t_{cal} by hand.
(c) Use the R output to test whether the mean starting salaries are different at $\alpha = 0.05$.

```
    F test to compare two variances

data:  uni_A and uni_B
F = 1.8666, num df = 9, denom df = 11, p-value = 0.3264
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.5202547 7.3023613
sample estimates:
ratio of variances
          1.866621


    Two Sample t-test

data:  uni_A and uni_B
```

```
t = 1.812, df = 20, p-value = 0.08502
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.3630677  5.1664011
sample estimates:
mean of x mean of y
 24.01000  21.60833
```

Formula Summary

Situation	Method	R
Paired data	paired t on d (Section 3.3)	`t.test(x1, x2, paired = TRUE)`
Independent, equal variances	pooled t: $s_p^2 = \dfrac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}$, $t_{\text{cal}} = \dfrac{\bar{x}_1-\bar{x}_2}{\sqrt{s_p^2(1/n_1+1/n_2)}}$, df $= n_1+n_2-2$	`t.test(x1, x2, var.equal = TRUE)`
Independent, unequal variances	Welch t	`t.test(x1, x2)`
CI (pooled)	$(\bar{x}_1-\bar{x}_2)\pm t_{1-\alpha/2,\,n_1+n_2-2}\,\text{SE}$	two-sided output

Exercises

3.5.1 An economist compares the monthly inflation expectations (%) reported by 14 firms in the North and 16 firms in the South (hypothetical data). She wants to know whether expectations are higher in the North. Three outputs are shown.

```
Output 1


    F test to compare two variances

data:  north and south
F = 8.0762, num df = 13, denom df = 15, p-value = 0.0002742
```

```
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
  2.761183 24.654312
sample estimates:
ratio of variances
          8.076196
```

Output 2

```
    Two Sample t-test

data:  north and south
t = 2.5738, df = 28, p-value = 0.007822
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 0.5712447       Inf
sample estimates:
mean of x mean of y
 8.828571  7.143750
```

Output 3

```
    Welch Two Sample t-test

data:  north and south
t = 2.4385, df = 15.809, p-value = 0.01347
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 0.4776595       Inf
sample estimates:
mean of x mean of y
 8.828571  7.143750
```

(a) Which of Output 2 and Output 3 should she use? Explain.
(b) Write the decision and interpretation at $\alpha = 0.05$.

3.5.2 For each study, say whether you would use the paired t test or a two-sample (pooled or Welch) t test.

(a) The rice yield of 15 farms is recorded before and after an irrigation project.
(b) The household income of 60 randomly chosen households in Chiang Mai is compared with that of 60 randomly chosen households in Bangkok.
(c) For each of 12 provinces, the unemployment rate of men is compared with that of women.

Solutions

Solution: Your Turn 3.5.1

(a) F test p-value $= 0.3264 \geq 0.05$: accept equal variances, so use the pooled t test.

(b) $s_p^2 = \dfrac{9(12.8677) + 11(6.8936)}{20} = 9.5819$, SE $= \sqrt{9.5819(1/10 + 1/12)} = 1.3254$, $t_{\text{cal}} = (24.01 - 21.6083)/1.3254 = 1.812$.

(c)

Step 1. $H_0 : \mu_A = \mu_B \quad H_1 : \mu_A \neq \mu_B$

Step 2. $\alpha = 0.05$

Step 3. $t_{\text{cal}} = 1.812$, df $= 20$

Step 4. p-value $= 0.0850$

Step 5. p-value $= 0.0850 \geq \alpha = 0.05$, so we accept H_0. That is, the mean starting salaries of graduates from the two universities are equal at the 0.05 level of significance.

Solution: Exercise 3.5.1

(a) In Output 1 the p-value $= 0.0003 < 0.05$, so the variances are not equal. Use the Welch test, Output 3. (Output 2 assumes equal variances.)

(b) $H_0 : \mu_N = \mu_S \quad H_1 : \mu_N > \mu_S$. From Output 3, $t_{\text{cal}} = 2.439$, df $= 15.81$, p-value $= 0.0135 < \alpha = 0.05$, so we reject H_0. That is, mean inflation expectations are higher in the North than in the South at the 0.05 level of significance.

Solution: Exercise 3.5.2

(a) Paired (same farms twice). (b) Two-sample: two independent samples; check the variances with the F test to choose pooled or Welch. (c) Paired (men and women are matched within each province).

4 Analysis of Variance

4.1 One-Way ANOVA: CRD

Learning Objectives

1. Explain why we use ANOVA, not many t tests, to compare three or more means.
2. Describe a completely randomized design (CRD) and the assumptions of one-way ANOVA.
3. Calculate the ANOVA table by hand for a small data set.
4. Read the output of `summary(aov(y ~ trt))` in R and write the conclusion.

Concept

A retail chain wants to compare **three** pricing strategies. Testing each pair with a t test needs three tests; with five strategies it needs ten. Each test has a 5% chance of a Type I error, so the chance of at least one false "significant" result grows quickly. **Analysis of variance (ANOVA)** tests all the means at once:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \qquad H_1 : \text{at least one } \mu_i \text{ is different}$$

In a **completely randomized design (CRD)** the treatments (here, pricing strategies) are assigned to experimental units (stores) completely at random.

The idea of ANOVA is to split the total variation in the data into two parts:

- variation **between** the treatment means (treatment sum of squares, SSTr)
- variation **within** the treatments (error sum of squares, SSE)

If the treatment means are really equal, the between-treatment variation is only random noise, about the same size as the within-treatment variation. The ratio of the two mean squares is an F statistic, as in Section 3.4.

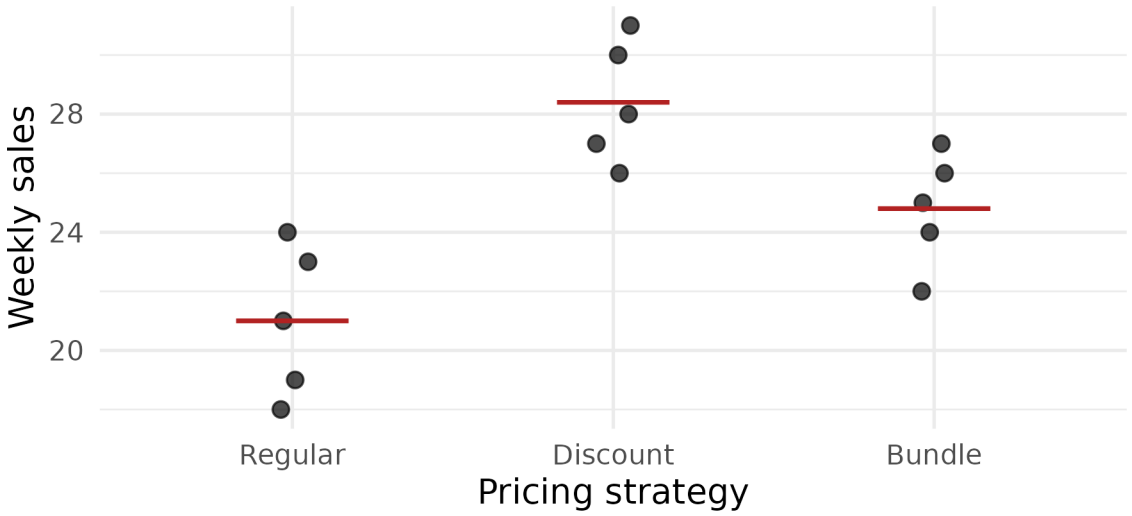


Figure 4.1: Weekly sales (hundred units) of 15 stores under three pricing strategies; lines show treatment means

Formulas and Conditions

Assumptions: independent random samples (units assigned at random); each population is normal; all populations have the same variance σ^2.

With k treatments, n_i observations in treatment i, $N = \sum n_i$, treatment totals T_i and grand total G:

Quantity	Formula
Correction factor	$\mathrm{CF} = G^2/N$
Total SS	$\mathrm{SST} = \sum y^2 - \mathrm{CF}$
Treatment SS	$\mathrm{SSTr} = \sum \dfrac{T_i^2}{n_i} - \mathrm{CF}$
Error SS	$\mathrm{SSE} = \mathrm{SST} - \mathrm{SSTr}$

ANOVA table

Source	df	SS	MS	F
Treatments	$k-1$	SSTr	$\mathrm{MSTr} = \dfrac{\mathrm{SSTr}}{k-1}$	$F_{\mathrm{cal}} = \dfrac{\mathrm{MSTr}}{\mathrm{MSE}}$
Error	$N-k$	SSE	$\mathrm{MSE} = \dfrac{\mathrm{SSE}}{N-k}$	
Total	$N-1$	SST		

Critical region: $F_{\mathrm{cal}} > F_{1-\alpha,\,k-1,\,N-k}$. The ANOVA F test is always **right-sided**: only a large between-treatment variation is evidence against H_0.

In R: `summary(aov(y ~ trt))`. The row named after the treatment is the treatment row and `Residuals` is the error row.

Worked Example

Worked Example 4.1.1. A retail chain randomly assigns three pricing strategies (Regular, Discount, Bundle) to 15 similar stores, 5 stores each, and records weekly sales in hundreds of units (hypothetical data):

Table 4.1: Weekly sales (hundred units) under three pricing strategies

Strategy	Sales	Total	Mean
Regular	21, 24, 19, 23, 18	105	21.0
Discount	28, 31, 26, 30, 27	142	28.4
Bundle	25, 22, 27, 24, 26	124	24.8

At $\alpha = 0.05$, test whether the mean weekly sales differ among the three strategies.

Solution

$k = 3$, $N = 15$, $G = 371$, $\sum y^2 = 9{,}371$.

$$\text{CF} = \frac{371^2}{15} = 9176.0667$$

$$\text{SST} = 9{,}371 - 9176.0667 = 194.9333$$

$$\text{SSTr} = \frac{105^2 + 142^2 + 124^2}{5} - 9176.0667 = 9313 - 9176.0667 = 136.9333$$

$$\text{SSE} = 194.9333 - 136.9333 = 58$$

Source	df	SS	MS	F
Strategy	2	136.9333	68.4667	14.166
Error	12	58	4.8333	
Total	14	194.9333		

Step 1: Hypotheses. $H_0 : \mu_R = \mu_D = \mu_B$ $\quad H_1 :$ at least one mean is different
Step 2: Level of significance. $\alpha = 0.05$
Step 3: Test statistic. $F_{\text{cal}} = \dfrac{68.4667}{4.8333} = 14.166$, df $= (2, 12)$
Step 4: p-value. From the R output below, p-value $= 0.0007$. (With the table: $F_{0.95,\,2,\,12} = 3.89$, critical region $F_{\text{cal}} > 3.89$.)
Step 5: Decision and interpretation. p-value $= 0.0007 < \alpha = 0.05$, so we reject H_0. That is, the mean weekly sales are not all equal across the three pricing strategies at the 0.05 level of significance. Section 4.2 shows how to find which strategies differ.

Reading R Output

```
            Df Sum Sq Mean Sq F value   Pr(>F)
strategy     2  136.9   68.47   14.17 0.000694 ***
Residuals   12   58.0    4.83
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Output	Meaning
`strategy` row	treatment: df $= k - 1$, SSTr, MSTr, F_{cal}, p-value
`Residuals` row	error: df $= N - k$, SSE, MSE
`Pr(>F)`	p-value
***, **, *	significance codes; we use the p-value itself

R does not print the Total row; add the other two rows to get it (df $= 14$, SS $= 194.9333$).

Your Turn

Your Turn 4.1.1

The household saving rate (% of income) of 4 randomly selected households in each of four regions is recorded (hypothetical data):

Table 4.2

Region	Saving rate	Total
North	12, 15, 11, 14	52
Northeast	9, 10, 8, 11	38
Central	16, 14, 17, 15	62

South	13, 11, 12, 14	50

$\sum y^2 = 2{,}648$. (a) Complete the ANOVA table by hand. (b) At $\alpha = 0.05$, test whether the mean saving rate differs among the regions, and check your table against the R output.

```
            Df Sum Sq Mean Sq F value   Pr(>F)
region       3  72.75  24.250   11.64 0.000725 ***
Residuals   12  25.00   2.083
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Formula Summary

	Formula
Hypotheses	$H_0 : \mu_1 = \cdots = \mu_k$; H_1: at least one μ_i different
Sums of squares	$\text{CF} = G^2/N$; $\text{SST} = \sum y^2 - \text{CF}$; $\text{SSTr} = \sum T_i^2/n_i - \text{CF}$; $\text{SSE} = \text{SST} - \text{SSTr}$
df	treatments $k-1$; error $N-k$; total $N-1$
Test statistic	$F_{\text{cal}} = \text{MSTr}/\text{MSE}$; reject H_0 if $F_{\text{cal}} > F_{1-\alpha,\,k-1,\,N-k}$
R	`summary(aov(y ~ trt))`

Exercises

4.1.1 Complete the ANOVA table for a CRD with 4 treatments and 6 units per treatment, and state the decision at $\alpha = 0.05$.

Source	df	SS	MS	F
Treatments	(a)	54.0	(c)	(e)
Error	(b)	80.0	(d)	
Total	23	134.0		

4.1.2 Explain why a researcher comparing five investment funds should use ANOVA rather than ten separate two-sample t tests.

Solutions

Solution: Your Turn 4.1.1

(a) $G = 202$, CF $= 202^2/16 = 2550.25$, SST $= 2648 - 2550.25 = 97.75$, SSTr $= (52^2 + 38^2 + 62^2 + 50^2)/4 - 2550.25 = 72.75$, SSE $= 25$.

Source	df	SS	MS	F
Region	3	72.75	24.25	11.640
Error	12	25	2.0833	
Total	15	97.75		

(b)

Step 1. $H_0 : \mu_N = \mu_{NE} = \mu_C = \mu_S \quad H_1 :$ at least one mean is different

Step 2. $\alpha = 0.05$

Step 3. $F_{\text{cal}} = 24.25/2.0833 = 11.640$, df $= (3, 12)$

Step 4. p-value $= 0.0007$ (table: $F_{0.95,\,3,\,12} = 3.49$)

Step 5. p-value $= 0.0007 < \alpha = 0.05$, so we reject H_0. That is, the mean saving rates are not all equal across the four regions at the 0.05 level of significance.

Solution: Exercise 4.1.1

(a) $k - 1 = 3$; (b) $N - k = 24 - 4 = 20$; (c) $54.0/3 = 18$; (d) $80.0/20 = 4$;
(b) $F_{\text{cal}} = 18/4 = 4.500$. $F_{0.95,\,3,\,20} = 3.1$; $4.500 > 3.1$, so F_{cal} falls in the critical region and we reject H_0: the treatment means are not all equal at the 0.05 level of significance.

Solution: Exercise 4.1.2

Each t test has a 5% chance of a Type I error. With ten tests the chance of at least one false significant result is much larger than 5%. ANOVA tests all five means in one test at $\alpha = 0.05$.

4.2 Multiple Comparisons (Tukey)

Learning Objectives

1. Explain when a multiple-comparison procedure is needed.
2. Read the output of `TukeyHSD()`: difference, confidence interval and adjusted p-value.
3. Decide which pairs of means differ and write the conclusion.

Concept

When ANOVA rejects H_0, we know that **at least one** mean is different, but not which ones. Comparing every pair with separate t tests would raise the overall chance of a Type I error, as Section 4.1 explained.

Tukey's honestly significant difference (HSD) method compares every pair of means while keeping the **family-wise** error rate (the chance of at least one Type I error among all the comparisons) at $\alpha = 0.05$. It is based on the Studentized range distribution (Table 14). In this course we read Tukey's results from R only.

> When to use Tukey
>
> Use Tukey only **after** the ANOVA F test rejects H_0. If ANOVA accepts H_0, the conclusion is that all the means are equal, and no pairwise comparison is needed.

Reading R Output

`TukeyHSD(model)` prints one row for each pair of treatments:

Column	Meaning
`diff`	difference between the two sample means, in the order of the row name (e.g. `Discount-Regular` $= \bar{x}_D - \bar{x}_R$)
`lwr`, `upr`	lower and upper limits of a 95% (family-wise) CI for the difference
`p adj`	adjusted p-value for $H_0 : \mu_i = \mu_j$

A pair is significantly different if `p adj` < 0.05; equivalently, if the CI from `lwr` to `upr` **does not contain 0**.

Worked Example

Worked Example 4.2.1. In Worked Example 4.1.1 the ANOVA rejected H_0 (p-value = 0.0007). The sample means are Regular 21, Discount 28.4 and Bundle 24.8. Use the Tukey output to find which pricing strategies differ at $\alpha = 0.05$.

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = sales ~ strategy, data = crd)

$strategy
                  diff         lwr        upr     p adj
Discount-Regular  7.4  3.69048613 11.1095139 0.0004927
Bundle-Regular    3.8  0.09048613  7.5095139 0.0445827
Bundle-Discount  -3.6 -7.30951387  0.1095139 0.0573998
```

Solution

Pair	diff	95% CI	p adj	Decision
Discount − Regular	7.4	(3.6905, 11.1095)	0.0005	< 0.05: different
Bundle − Regular	3.8	(0.0905, 7.5095)	0.0446	< 0.05: different
Bundle − Discount	-3.6	(-7.3095, 0.1095)	0.0574	≥ 0.05: not different

At the 0.05 level of significance:

- mean sales with the Discount strategy are higher than with the Regular strategy;
- mean sales with the Bundle strategy are higher than with the Regular strategy;
- mean sales with the Discount and Bundle strategies are equal.

Both promotional strategies beat the regular price, but there is no significant difference between them.

Your Turn

Your Turn 4.2.1

In Your Turn 4.1.1 the ANOVA showed that the mean saving rates differ among the four regions. Use the output below to decide which pairs of regions differ at $\alpha = 0.05$, and summarize the result in one or two sentences.

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = saving ~ region, data = sav)

$region
                    diff         lwr         upr     p adj
Northeast-North   -3.5 -6.53012202 -0.46987798 0.0224076
Central-North      2.5 -0.53012202  5.53012202 0.1199630
South-North       -0.5 -3.53012202  2.53012202 0.9598670
Central-Northeast  6.0  2.96987798  9.03012202 0.0003771
South-Northeast    3.0 -0.03012202  6.03012202 0.0526110
South-Central     -3.0 -6.03012202  0.03012202 0.0526110
```

Formula Summary

When	after ANOVA rejects H_0
R	`TukeyHSD(model)` (CRD); `TukeyHSD(model, "trt")` (RCBD, Section 4.3)
Decision for a pair	different if `p adj` < 0.05, equivalently if (`lwr`, `upr`) does not contain 0
Direction	sign of `diff` (row `A-B` means $\bar{x}_A - \bar{x}_B$)

Exercises

4.2.1 A Tukey output for three loan products contains the row

```
                diff       lwr       upr     p adj
Car-Personal  -1.85  -3.12000  -0.58000 0.0041000
```

Interpret this row.

4.2.2 An ANOVA comparing four advertising channels gives p-value = 0.2840. A student then runs `TukeyHSD()` and finds one pair with p adj = 0.0470. What should the conclusion be?

Solutions

Solution: Your Turn 4.2.1

Pairs with p adj < 0.05 (CI does not contain 0): Northeast − North (p adj = 0.0224); Central − Northeast (p adj = 0.0004). All other pairs are not significantly different.

That is, at the 0.05 level of significance the mean saving rate in the Northeast is lower than in the North and lower than in the Central region; the other pairs of regions have equal mean saving rates.

Solution: Exercise 4.2.1

The mean for car loans is 1.85 lower than for personal loans ($\bar{x}_{\text{Car}} - \bar{x}_{\text{Personal}} = -1.85$). The 95% CI (-3.12, -0.58) does not contain 0 and p adj = 0.0041 < 0.05, so the mean for car loans is lower than for personal loans at the 0.05 level of significance.

Solution: Exercise 4.2.2

The ANOVA accepts H_0 ($0.2840 \geq 0.05$), so the conclusion is that the mean outcomes of the four channels are equal. Tukey's method should not be used after ANOVA accepts H_0.

4.3 Randomized Complete Block Design (RCBD)

Learning Objectives

1. Explain what a block is and why blocking makes a comparison more precise.
2. Describe a randomized complete block design (RCBD).
3. Calculate the RCBD ANOVA table and read `summary(aov(y ~ trt + block))`.
4. Follow a significant result with `TukeyHSD(model, "trt")`.

Concept

Suppose the retail chain from Section 4.1 has stores in five provinces. Sales in Chiang Mai are much higher than in Nan, whatever the pricing strategy. In a CRD this province-to-province variation ends up in the error term, makes MSE large and hides the differences between strategies.

A **block** is a group of similar experimental units. In a **randomized complete block design (RCBD)**:

- every treatment appears **once in every block** (the design is "complete");
- within each block, the treatments are assigned to units **at random**.

Here each province is a block: in each province, three stores are randomly assigned to the three strategies. Comparing strategies **within** each province removes the province effect, and ANOVA separates the variation into three parts: treatments, blocks and error.

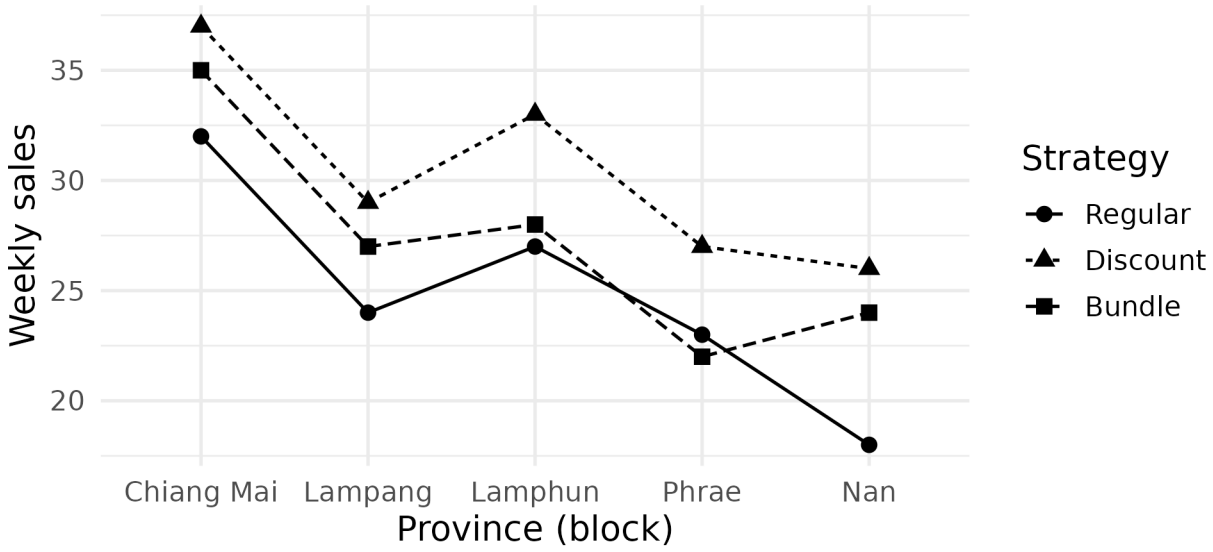


Figure 4.2: Weekly sales by strategy in each province. Provinces differ a lot, but within every province the Discount strategy sells most.

Formulas and Conditions

Assumptions: within each block the treatments are assigned at random; the errors are independent and normal with constant variance; no interaction between treatments and blocks.

With t treatments, b blocks, $N = tb$, treatment totals T_i, block totals B_j and grand total G:

Quantity	Formula
Correction factor	$\mathrm{CF} = G^2/N$
Total SS	$\mathrm{SST} = \sum y^2 - \mathrm{CF}$
Treatment SS	$\mathrm{SSTr} = \dfrac{\sum T_i^2}{b} - \mathrm{CF}$
Block SS	$\mathrm{SSB} = \dfrac{\sum B_j^2}{t} - \mathrm{CF}$
Error SS	$\mathrm{SSE} = \mathrm{SST} - \mathrm{SSTr} - \mathrm{SSB}$

Source	df	SS	MS	F
Treatments	$t-1$	SSTr	MSTr	MSTr/MSE
Blocks	$b-1$	SSB	MSB	MSB/MSE
Error	$(t-1)(b-1)$	SSE	MSE	
Total	$tb-1$	SST		

Test for treatments: $H_0 : \mu_1 = \cdots = \mu_t$; reject H_0 if $F_{\text{cal}} > F_{1-\alpha,\, t-1,\, (t-1)(b-1)}$. The block row is printed by R, but our question is about the treatments.

In R: `summary(aov(y ~ trt + block))`, then `TukeyHSD(model, "trt")` to compare the treatments only.

Worked Example

Worked Example 4.3.1. In each of five provinces, three stores are randomly assigned to the Regular, Discount and Bundle pricing strategies. Weekly sales (hundred units) are recorded (hypothetical data):

Table 4.3: Weekly sales (hundred units): strategies (rows) in five provinces (blocks)

	Chiang Mai	Lampang	Lamphun	Phrae	Nan	Total
Regular	32	24	27	23	18	124
Discount	37	29	33	27	26	152
Bundle	35	27	28	22	24	136
Total	104	80	88	72	68	412

At $\alpha = 0.05$, test whether the mean weekly sales differ among the strategies. If they do, find which strategies differ.

Solution

$t = 3$, $b = 5$, $N = 15$, $G = 412$, $\sum y^2 = 11{,}684$.

$$\text{CF} = \frac{412^2}{15} = 11316.2667, \qquad \text{SST} = 11,684 - 11316.2667 = 367.7333$$

$$\text{SSTr} = \frac{124^2 + 152^2 + 136^2}{5} - 11316.2667 = 78.9333$$

$$\text{SSB} = \frac{104^2 + 80^2 + 88^2 + 72^2 + 68^2}{3} - 11316.2667 = 273.0666$$

$$\text{SSE} = 367.7333 - 78.9333 - 273.0666 = 15.7334$$

Source	df	SS	MS	F
Strategy	2	78.9333	39.4667	20.067
Province	4	273.0666	68.2667	34.711
Error	8	15.7334	1.9667	
Total	14	367.7333		

Step 1: Hypotheses. $H_0 : \mu_R = \mu_D = \mu_B$ H_1 : at least one mean is different
Step 2: Level of significance. $\alpha = 0.05$
Step 3: Test statistic. $F_{\text{cal}} = 39.4667/1.9667 = 20.067$, df = (2, 8)
Step 4: p-value. From the R output below, p-value = 0.0008. (With the table: $F_{0.95,2,8} = 4.46$.)
Step 5: Decision and interpretation. p-value = 0.0008 $< \alpha = 0.05$, so we reject H_0. That is, the mean weekly sales are not all equal across the three pricing strategies at the 0.05 level of significance.
Which strategies differ? From `TukeyHSD(model, "strategy")` below: Discount − Regular (p adj = 0.0006) and Bundle − Discount (p adj = 0.0169) are significant; Bundle − Regular (p adj = 0.0625) is not. At the 0.05 level of significance the Discount strategy has higher mean sales than both the Regular and the Bundle strategies, and the Regular and Bundle strategies have equal mean sales.

Reading R Output

```
            Df Sum Sq Mean Sq F value   Pr(>F)
strategy     2  78.93   39.47   20.07 0.000763 ***
province     4 273.07   68.27   34.71 4.21e-05 ***
Residuals    8  15.73    1.97
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = sales ~ strategy + province, data = rb)

$strategy
                  diff        lwr        upr     p adj
Discount-Regular  5.6  3.0656117  8.1343883 0.0005906
Bundle-Regular    2.4 -0.1343883  4.9343883 0.0624855
Bundle-Discount  -3.2 -5.7343883 -0.6656117 0.0168916
```

The `strategy` row is the treatment test; the `province` row shows that the blocks differ a lot, which is why blocking helps.

What if we ignore the blocks?

Analysing the same data as a CRD, `summary(aov(sales ~ strategy))`, puts the province variation into the error term:

```
            Df Sum Sq Mean Sq F value Pr(>F)
strategy     2  78.93   39.47    1.64  0.235
Residuals   12 288.80   24.07
```

The treatment SS is the same (78.9333), but the error SS grows from 15.7334 to 288.8, the p-value becomes 0.2346 and the difference between the strategies is no longer detected. The design and the analysis must match.

Your Turn

Your Turn 4.3.1

A regulator records the interest rate (%) of three loan products (Personal, Car, Home) at six banks (hypothetical data). Each bank offers all three products, so the banks are blocks.

Table 4.4

	Bank 1	Bank 2	Bank 3	Bank 4	Bank 5	Bank 6
Personal	8.4	7.9	9.1	8.8	7.6	9.5
Car	7.4	6.3	8.6	7.9	7.1	8.0
Home	7.0	6.4	7.9	8.1	6.0	8.4

(a) At $\alpha = 0.05$, test whether the mean interest rates of the three products differ.
(b) If they do, find which products differ.

```
            Df Sum Sq Mean Sq F value   Pr(>F)
product      2  5.250  2.6250   23.51 0.000166 ***
bank         5  9.513  1.9027   17.04 0.000131 ***
Residuals   10  1.117  0.1117
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  Tukey multiple comparisons of means
    95% family-wise confidence level
```

```
Fit: aov(formula = rate ~ product + bank, data = loans)

$product
                diff        lwr        upr     p adj
Car-Personal   -1.00 -1.5288798 -0.4711202 0.0010846
Home-Personal -1.25 -1.7788798 -0.7211202 0.0001896
Home-Car       -0.25 -0.7788798  0.2788798 0.4288716
```

Formula Summary

	Formula
Sums of squares	SSTr $= \sum T_i^2/b - \text{CF}$; SSB $= \sum B_j^2/t - \text{CF}$; SSE $=$ SST $-$ SSTr $-$ SSB
df	treatments $t-1$; blocks $b-1$; error $(t-1)(b-1)$; total $tb-1$
Test for treatments	$F_{\text{cal}} = \text{MSTr/MSE}$, df $= (t-1,\ (t-1)(b-1))$
R	`summary(aov(y ~ trt + block))`; `TukeyHSD(model, "trt")`

Exercises

4.3.1 For each study, say whether it is a CRD or an RCBD, and name the treatments and (if any) the blocks.

(a) Twenty shops are randomly assigned to four store layouts, five shops each.
(b) Four store layouts are each tried once in each of five shopping malls, with the layout randomly assigned to shops within each mall.

4.3.2 An RCBD has 4 treatments and 6 blocks, with SST = 120, SSTr = 36 and SSB = 54. Complete the ANOVA table and test the treatments at $\alpha = 0.05$.

Solutions

Solution: Your Turn 4.3.1

(a)

Step 1. $H_0 : \mu_P = \mu_C = \mu_H \quad H_1 :$ at least one mean is different

Step 2. $\alpha = 0.05$

Step 3. $F_{\text{cal}} = 23.507$, df = (2, 10)

Step 4. p-value = 0.0002

Step 5. p-value = $0.0002 < \alpha = 0.05$, so we reject H_0. That is, the mean interest rates are not all equal across the three loan products at the 0.05 level of significance.

(b) Car − Personal (p adj = 0.0011) and Home − Personal (p adj = 0.0002) are significant; Home − Car (p adj = 0.4289) is not. At the 0.05 level of significance, personal loans have a higher mean interest rate than car loans and home loans, and car and home loans have equal mean rates.

Solution: Exercise 4.3.1

(a) CRD; treatments: the four layouts; no blocks.
(b) RCBD; treatments: the four layouts; blocks: the five malls.

Solution: Exercise 4.3.2

SSE = 120 − 36 − 54 = 30; df: treatments 3, blocks 5, error 15, total 23. MSTr = 36/3 = 12, MSB = 54/5 = 10.8, MSE = 30/15 = 2. $F_{\text{cal}} = 12/2 = 6.000$ for treatments. $F_{0.95,\,3,\,15} = 3.29$; $6.000 > 3.29$, so F_{cal} falls in the critical region and we reject H_0: the treatment means are not all equal at the 0.05 level of significance.

Summary: Choosing a Method

Which Method? A Flowchart

Every problem in this book answers the same three questions:

1. **Which parameter?** A mean, a proportion or a variance.
2. **How many groups?** One, two, or three or more.
3. **What is special about the data?** Is σ known? Are the samples paired? Are the variances equal? Are there blocks?

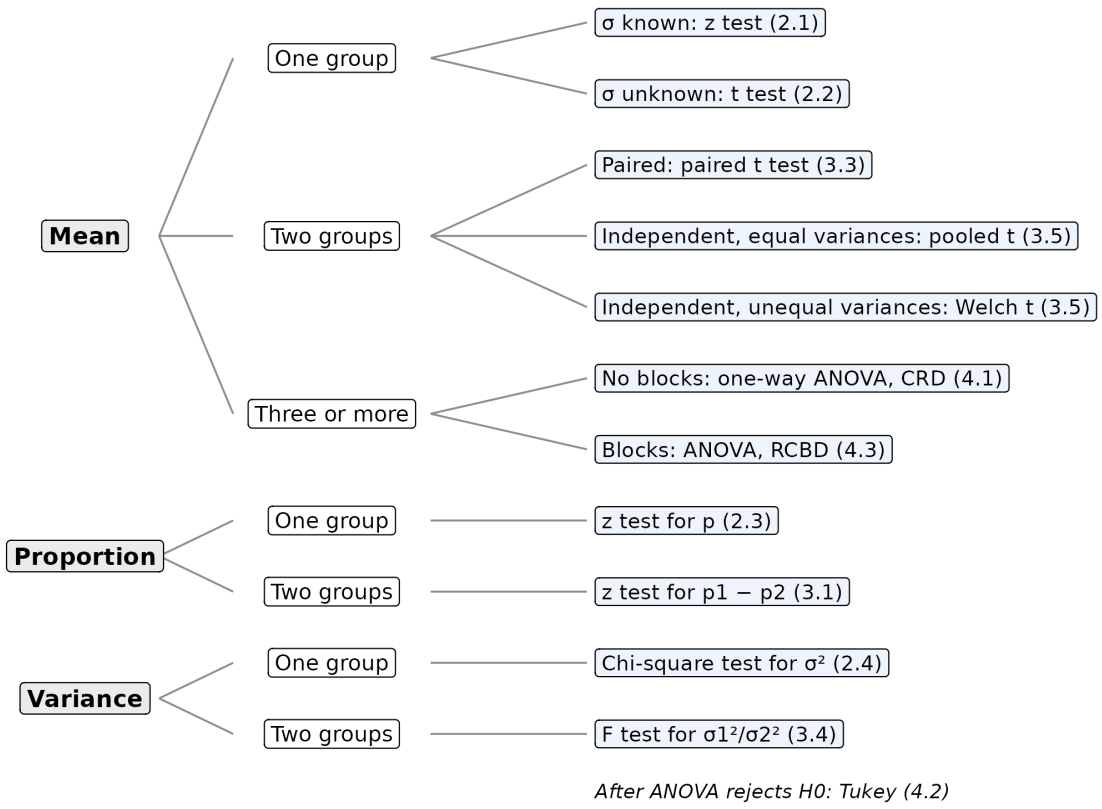


Figure 4.3: Choosing a method. Numbers in brackets are the sections.

Method	Test statistic	Distribution	R
Mean, σ known	$z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$	z	–
Mean, σ unknown	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	t, $n-1$	`t.test(x, mu = )`
Proportion	$z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$	z	–
Variance	$\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$	χ^2, $n-1$	–
Two proportions	$z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\bar{p}(1-\bar{p})(1/n_1 + 1/n_2)}}$	z	–
Paired means	$t = \dfrac{\bar{d}}{s_d/\sqrt{n}}$	t, $n-1$	`t.test(x1, x2, paired = TRUE)`
Two variances	$F = s_1^2/s_2^2$	F, (n_1-1, n_2-1)	`var.test(x1, x2)`
Two means, pooled	$t = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2(1/n_1 + 1/n_2)}}$	t, n_1+n_2-2	`t.test(x1, x2, var.equal = TRUE)`
Two means, Welch	–	t	`t.test(x1, x2)`
ANOVA (CRD)	$F = \text{MSTr}/\text{MSE}$	F, $(k-1, N-k)$	`summary(aov(y ~ trt))`
ANOVA (RCBD)	$F = \text{MSTr}/\text{MSE}$	F, $(t-1, (t-1)(b-1))$	`summary(aov(y ~ trt + block))`
Tukey	–	–	`TukeyHSD(model)`

Which Method? Practice

For each study, name the parameter(s), the method and the section.

(a) The Bank of Thailand wants to know whether the share of households with credit-card debt has risen above 35%.
(b) A researcher compares the mean wage of 40 workers in Chiang Mai with the mean wage of 35 workers in Khon Kaen.
(c) The same 25 firms report their profits before and after a tax change.
(d) An analyst asks whether the daily returns of a stock index have become more volatile than a standard deviation of 1.2%.
(e) Four fertilizer subsidy schemes are each tried in five districts, with the scheme randomly assigned to villages within each district.
(f) A survey compares the proportion of men and of women who own a business.

Mixed Problems

All figures are hypothetical. Use $\alpha = 0.05$ and the five steps.

M1. In a survey of 500 workers, 210 have a second job. Test whether more than 40% of workers have a second job. Use the critical region.

M2. The monthly profit (thousand baht) of 9 cafés in the old city has $\bar{x} = 52.4$, $s = 6.1$; that of 13 cafés near the university has $\bar{x} = 47.8$, $s = 5.4$. (a) Use the F test with the critical region to check the variances. (b) Test whether the mean profits are different, using the critical region.

M3. A central bank wants the standard deviation of the monthly inflation rate to be less than 1 percentage point. Over 15 months, $s = 0.8$. Assume normality and test whether the target is met, using the critical region.

M4. Eight households installed solar panels. Their monthly electricity bills (baht) before and after installation were recorded. Did the mean bill decrease?

```
    Paired t-test

data:  bill_after and bill_before
t = -4.4458, df = 7, p-value = 0.001493
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
      -Inf -216.6301
sample estimates:
mean difference
         -377.5
```

M5. Twenty stores are randomly assigned to four advertising channels (TV, Online, Radio, Print), five stores each, and weekly sales are recorded. (a) Test whether mean sales differ among the channels. (b) If they do, which channels differ?

```
            Df Sum Sq Mean Sq F value   Pr(>F)
channel      3    466  155.33   13.08 0.000142 ***
Residuals   16    190   11.87
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = sales ~ channel, data = ads)

$channel
                diff        lwr       upr     p adj
Online-TV       1.52  -4.714923  7.754923 0.8965475
Radio-TV       -9.02 -15.254923 -2.785077 0.0038601
Print-TV       -8.64 -14.874923 -2.405077 0.0055055
Radio-Online -10.54 -16.774923 -4.305077 0.0009438
Print-Online -10.16 -16.394923 -3.925077 0.0013384
Print-Radio    0.38  -5.854923  6.614923 0.9980348
```

M6. The mean household size in a district is estimated from 20 households: $\bar{x} = 3.42$, $s = 0.58$. Construct a 95% confidence interval for the mean household size.

Writing a Full Conclusion

A full conclusion has a **decision** and an **interpretation** in the words of the problem:

> **Decision.** [statistic or p-value] … so we [reject / accept] H_0. **Interpretation.** That is, … at the 0.05 level of significance.

Checklist:

- If you reject H_0, write H_1 as a sentence; if you accept H_0, write H_0 as a sentence.
- Name the population and the variable (not just "the mean").
- For a two-sided test that rejects H_0, add the direction from the sample statistics.
- For ANOVA, "not all equal"; then use Tukey to say which ones differ.

Write the decision and the interpretation for each result.

(a) $H_0 : \mu = 15000$, $H_1 : \mu \neq 15000$ for the mean monthly rent (baht) in a district; $t_{\text{cal}} = -2.514$, critical region $|t_{\text{cal}}| > 2.064$, $\bar{x} = 13{,}850$.

(b) $H_0 : p_1 = p_2$, $H_1 : p_1 > p_2$ for the proportions of urban (1) and rural (2) households using mobile banking; p-value $= 0.0832$.

(c) One-way ANOVA comparing mean loan sizes at three types of lender; p-value $= 0.0007$.

Solutions

Solution: Which Method? Practice

(a) One proportion, z test for p (2.3), $H_1 : p > 0.35$.

(b) Two independent means: F test for the variances first, then pooled or Welch t (3.4, 3.5).

(c) Paired means: paired t test on the differences (3.3).

(d) One variance: chi-square test, $H_1 : \sigma^2 > 1.44$ (2.4).

(e) Four means with blocks (districts): RCBD ANOVA (4.3), then Tukey if H_0 is rejected.

(f) Two proportions: z test for $p_1 - p_2$ (3.1).

Solution: M1

One proportion (2.3). $\hat{p} = 210/500 = 0.42$.

Step 1. $H_0 : p = 0.40 \quad H_1 : p > 0.40$

Step 2. $\alpha = 0.05$

Step 3. $\sqrt{0.40(0.60)/500} = 0.0219$; $z_{\text{cal}} = (0.42 - 0.40)/0.0219 = 0.913$

Step 4. Critical region: $z_{\text{cal}} > 1.645$.

Step 5. $z_{\text{cal}} = 0.913$ does not fall in the critical region ($0.913 < 1.645$), so we accept H_0. That is, the proportion of workers with a second job is equal to 0.40 at the 0.05 level of significance.

Solution: M2

(a) $H_0 : \sigma_1^2 = \sigma_2^2$ vs $H_1 : \sigma_1^2 \neq \sigma_2^2$. $F_{\text{cal}} = 37.21/29.16 = 1.276$, df = (8, 12). Critical region: $F_{\text{cal}} < 1/F_{0.975,\,12,\,8} = 0.2381$ or $F_{\text{cal}} > F_{0.975,\,8,\,12} = 3.51$. F_{cal} does not fall in the critical region, so we accept H_0: the variances are equal. Use the pooled t test.

(b)

Step 1. $H_0 : \mu_1 = \mu_2 \quad H_1 : \mu_1 \neq \mu_2$

Step 2. $\alpha = 0.05$

Step 3. $s_p^2 = [8(37.21) + 12(29.16)]/20 = 32.38$; SE $= \sqrt{32.38(1/9 + 1/13)} = 2.4675$; $t_{\text{cal}} = (52.4 - 47.8)/2.4675 = 1.864$, df = 20

Step 4. Critical region: $t_{\text{cal}} < -2.086$ or $t_{\text{cal}} > 2.086$.

Step 5. $t_{\text{cal}} = 1.864$ does not fall in the critical region ($-2.086 < 1.864 < 2.086$), so we accept H_0. That is, the mean monthly profits of cafés in the two areas are equal at the 0.05 level of significance.

Solution: M3

One variance (2.4).

Step 1. $H_0 : \sigma^2 = 1 \quad H_1 : \sigma^2 < 1$

Step 2. $\alpha = 0.05$

Step 3. $\chi^2_{\text{cal}} = 14(0.64)/1 = 8.960$, df = 14

Step 4. $\chi^2_{0.05,\,14} = 6.57$. Critical region: $\chi^2_{\text{cal}} < 6.57$.

Step 5. $\chi^2_{\text{cal}} = 8.960$ does not fall in the critical region ($8.960 > 6.57$), so we accept H_0. That is, the standard deviation of monthly inflation is equal to 1 percentage point at the 0.05 level of significance; the data do not show that the target is met.

Solution: M4

Paired means (3.3), d = after − before.

Step 1. $H_0 : \mu_d = 0 \quad H_1 : \mu_d < 0$

Step 2. $\alpha = 0.05$

Step 3. $t_{\text{cal}} = -4.446$, df = 7

Step 4. p-value = 0.0015

Step 5. p-value $= 0.0015 < \alpha = 0.05$, so we reject H_0. That is, the mean monthly electricity bill decreased after installing solar panels at the 0.05 level of significance.

Solution: M5

(a) One-way ANOVA, CRD (4.1).

Step 1. $H_0 : \mu_{TV} = \mu_O = \mu_R = \mu_P \quad H_1 :$ at least one mean is different

Step 2. $\alpha = 0.05$

Step 3. $F_{\text{cal}} = 13.082$, df = (3, 16)

Step 4. p-value = 0.0001

Step 5. p-value $= 0.0001 < \alpha = 0.05$, so we reject H_0. That is, mean weekly sales are not all equal across the four channels at the 0.05 level of significance.

(b) Pairs with p adj < 0.05: Radio − TV, Print − TV, Radio − Online, Print − Online. TV and Online have equal mean sales, Radio and Print have equal mean sales, and both TV and Online have higher mean sales than both Radio and Print.

Solution: M6

One mean, σ unknown (2.2). $t_{0.975,\,19} = 2.093$, $s/\sqrt{n} = 0.1297$; $3.42 \pm 2.093(0.1297) = 3.42 \pm 0.2715$. We are 95% confident that the mean household size is between 3.15 and 3.69 persons.

Solution: Writing a Full Conclusion

(a) $t_{\text{cal}} = -2.514$ falls in the critical region ($-2.514 < -2.064$), so we reject H_0. That is, the mean monthly rent in the district is different from 15,000 baht at the 0.05 level of significance; since $\bar{x} = 13{,}850$, it is lower than 15,000 baht.

(b) p-value $= 0.0832 \geq \alpha = 0.05$, so we accept H_0. That is, the proportions of urban and rural households using mobile banking are equal at the 0.05 level of significance.

(c) p-value $= 0.0007 < \alpha = 0.05$, so we reject H_0. That is, the mean loan sizes are not all equal across the three types of lender at the 0.05 level of significance. (Use Tukey's method to find which types differ.)